NETLIST → LIBRARIES → TECH FILES → FLOORPLAN → PLACEMENT → OPTIMIZATION → SIGNOFF

- Gate Level Netlist
- Libraries
- Constraints

- LEF
- DEF

- DRC
- LVS
- Antenna
- Timing

PD PLACEMENT

PD PLACEMENT

Optimizing Cell Locations for Performance,
Congestion Reduction and Timing Closure.

IN → OUT

Good Placement → Better Performance

PLACEMENT OBJECTIVES

- High Performance
- Low Congestion
- Timing Closure
- Better QoR
- Power Efficiency
- Area Efficiency
- DFM Compliance

KEY ENGINES

TIMING ENGINE
- STA
- Slack Analysis
- Path Optimization

CONGESTION ENGINE
- Congestion Map
- Hotspot Detection
- Resource Balance

PLACEMENT ENGINE
- Global Placement
- Legalization
- Optimization

POWER ENGINE
- Power Estimation
- IR Drop Analysis
- Power Optimization

DRC / LVS ENGINE
- Rule Checking
- LVS Verification
- Clean Signoff

QOR ENGINE
- Area / Timing / Power
- Multi-Corner Analysis
- QoR Reporting

CLK

Author :
Konda Tharun Vivekananda

VDD

"Great Placement + Smart Optimization = Exceptional Chip.

Your time and attention are
greatly appreciated.

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PLACEMENT

1. Placement – Introduction

Placement is the process of deciding the physical locations (X,Y) of all movable standard cells and macros inside the core area after floorplan and power planning. It converts the logical netlist into a physical representation while optimizing timing, congestion, wirelength and routability.

2. Why Needed ?

- Determines the initial wirelength and congestion.
- Major impact on timing, area and power.
- Provides good starting point for CTS and Routing.
- Improves routability and reduces iterations.

3. Advantages

- Reduced wirelength → better timing & power.
- Lower congestion → easy routing.
- Better timing closure.
- Efficient area utilization.
- Fewer downstream iterations.

4. Applications

- ASICs, SoCs, CPUs, GPUs, DSPs.
- High performance and low power designs.
- Memory chips and mixed signal ICs.
- Any digital design after synthesis.

5. Input Files Required

- Gate Level Netlist : .v / .v.gz
- Floorplan (DEF) : .def
- Technology LEF : .lef
- Timing Libraries : .lib
- Constraints : .sdc
- Power Nets : UPF / DEF
- Placement Rules / Guides : .tcl / .conf (optional)

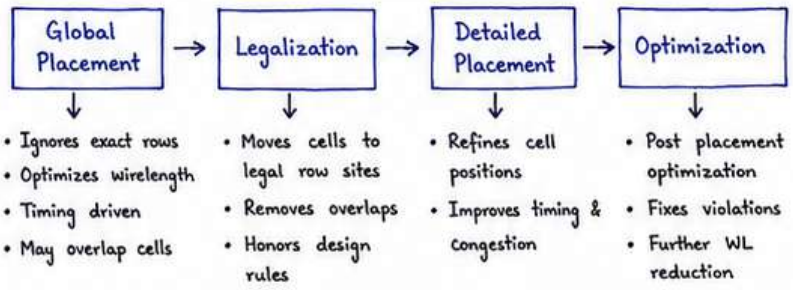
Key Point

Good placement reduces wirelength and congestion, improves timing and makes routing easy.

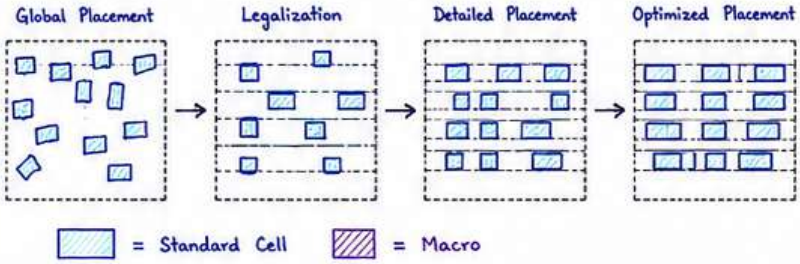
6. Output Files Generated

- Placement DEF : placed.def
- Placement Database : .pl / .odb
- Placement Log : .log
- Cell Location Report : .rpt
- Congestion Report : .rpt
- Wirelength Report : .rpt
- Timing (Pre-CTS) Report : .rpt

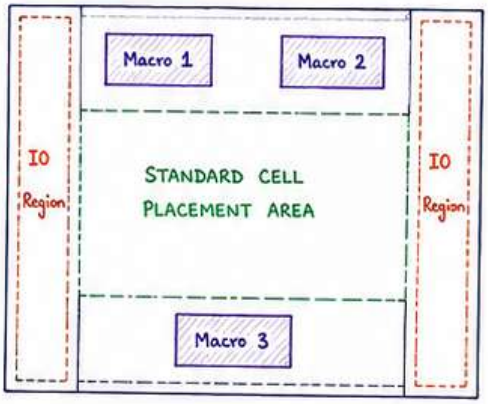
7. Placement Flow



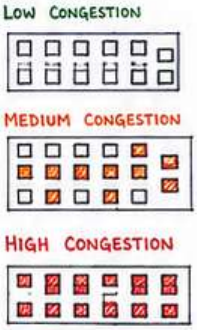
8. Placement Process (Visual)



9. Placement Region (Example)



10. Congestion (Idea)



→ High routing demand
↓
Congestion
↓
Detours in routing
↓
Timing degradation
↓
Routing failure

11. Placement Goals

- ✓ Minimize wirelength
- ✓ Improve timing (setup/hold)
- ✓ Reduce congestion
- ✓ Balance area utilization
- ✓ Minimize power
- ✓ Meet design rules
- ✓ Good routability

12. Important Metrics

Metric	Meaning	Target / Goal
Wirelength (WL)	Total length of nets	Minimize
Congestion	Routing resource demand	< 1.0 (ideal)
Cell Density	Cells per unit area	Balanced
Timing Slack	Setup / Hold slack	> 0 ns
HPWL / nPO	Half-Perimeter WL per net	Minimize

13. Practical Observation

→ Early good placement reduces timing and routing issues later.

→ Congestion hot-spots should be removed during placement.

→ Timing driven placement gives better slack.

→ Iterative placement is key to timing closure.

14. Interview / Viva Points

Q. What is placement in Physical Design ?

→ Placement decides the physical locations of cells to optimize timing, wirelength, congestion and routability.

Q. Why is placement important ?

→ It strongly affects timing, power, routing congestion and overall PPA of the design.

15. Summary / Revision Points

1. Placement assigns physical locations to cells.
2. It includes global placement, legalization, detailed placement and optimization.
3. Goals: minimize WL, reduce congestion, improve timing and ensure routability.
4. Good placement is critical for timing closure.
5. Outputs of placement are DEF, reports, logs and databases for CTS and routing.

PLACEMENT

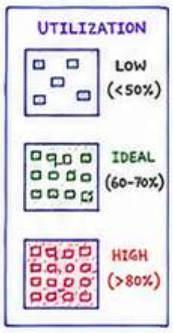
KEY TERMINOLOGIES

1. UTILIZATION

- It is the percentage of core area occupied by standard cells.
- Higher utilization → better area efficiency.
- Too high utilization → routing congestion, timing degradation.
- Too low utilization → wasted area.

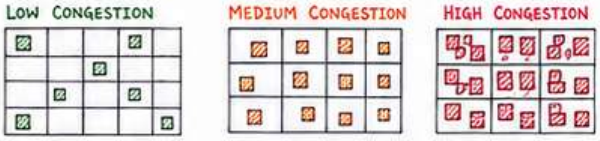
Formula : Utilization (%) = (Total cell area) / (Placeable area) × 100

Typical Range : 60% – 75%
Ideal : ~ 65%



2. CONGESTION

- It is the measure of extra routing demand over available routing resources in a region.
- High congestion → detours, timing degradation, routing failures.
- Measured during placement and after global routing.



Congestion (%) = (Total Routing Demand) / (Total Routing Capacity) × 100

3. OVERFLOW

- Overflow is the percentage of cells that cannot be placed in available space (during global placement).
- It indicates how many cells are outside the legal placement area.
- Goal is to reduce overflow to 0 before legalization.

Overflow (%) = (Unplaced cell area) / (Total cell area) × 100

4. WNS (Worst Negative Slack)

- WNS is the most negative slack among all timing paths.
- It represents the worst timing violation in the design.
- Negative WNS → timing not closed.
- Positive WNS → timing met.

WNS = min (Slack of all timing paths)

WNS < 0 → Timing Violation
WNS > 0 → Timing Met

Practical Note

Placement directly impacts WNS by affecting wirelength, cell locations and congestion.

5. TNS (Total Negative Slack)

- TNS is the sum of negative slacks of all failing paths.
- It shows the total amount of timing violation in the design.
- More negative TNS → higher violation.

TNS = ∑ (Slack of all paths where Slack < 0)

Example

Path	Slack (ns)
P_1	−0.12
P_2	−0.25
P_3	+0.10
P_4	−0.05

WNS = −0.25 ns
TNS = −0.12 −0.25 −0.05 = −0.42 ns

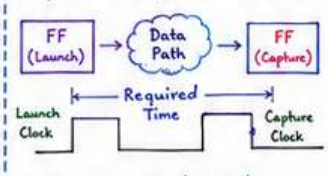
6. TIMING SLACK

- Slack is the difference between required time and arrival time.
- Positive Slack → path meets timing.
- Negative Slack → timing violation.
- It is the most important timing metric.

Slack = Required Time − Arrival Time

Setup Slack

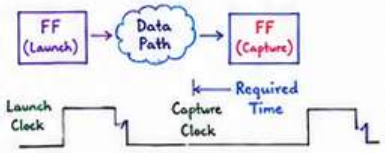
Setup Slack = Required Time − Data Arrival Time



Slack > 0 → OK (GREEN)
Slack < 0 → VIOLATION (RED)

Hold Slack

Hold Slack = Data Arrival Time − Required Time

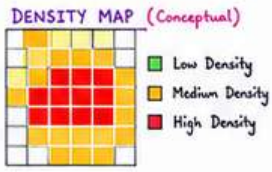


Slack > 0 → OK (GREEN)
Slack < 0 → VIOLATION (RED)

7. DENSITY

- Density is the distribution of cell area in a region.
- It is usually expressed as cell area per unit area.
- Uniform density → better routability.
- High density in small region → congestion.

Density = Total cell area in region / Area of region (%)



8. QUICK REFERENCE TABLE

Metric	Meaning	Formula / Definition	Good Indication
Utilization (%)	% of core area occupied by cells	(Total cell area / Placeable area) × 100	60% – 75% (Ideal ~65%)
Congestion (%)	Extra routing demand over available capacity	(Routing Demand / Routing Capacity) × 100	Low is Good (<100%)
Overflow (%)	% of cells outside legal placement area	(Unplaced cell area / Total cell area) × 100	Goal is 0%
WNS (ns)	Worst (most negative) slack	min (Slack of all paths)	> 0 → Good < 0 → Violation
TNS (ns)	Total of negative slacks of failing paths	∑ (Slack of paths where Slack < 0)	Closer to 0 (Good)
Timing Slack (ns)	Difference between required and arrival time	Slack = Required − Arrival	> 0 → Good < 0 → Violation
Density (%)	Cell area per unit area in a region	(Cell area in region / Region area) × 100	Uniform & Moderate is Good

9. PRACTICAL OBSERVATIONS

→ Good placement balances utilization and congestion.
→ High utilization may cause high congestion.
→ Overflow must be reduced to zero before legalization.
→ Better placement improves timing (WNS / TNS).
→ Density uniformity helps in better routability.
→ Metrics are continuously monitored during optimization.

10. INTERVIEW / VIVA Q&A

Q. What is utilization?
→ It is the % of core area occupied by cells.
Q. What is congestion?
→ Extra routing demand over available routing capacity.
Q. What is WNS?
→ Worst (most negative) slack among all paths.
Q. What is TNS?
→ Sum of negative slacks of all failing paths.

11. SUMMARY POINTS

1. Utilization shows how efficiently area is used.
2. Congestion affects routability and timing.
3. Overflow must be 0 before legalization.
4. WNS indicates worst timing violation.
5. TNS shows total amount of violation.
6. Timing slack determines if path meets timing.
7. Density uniformity improves routing.
8. These metrics guide placement optimization.

PLACEMENT

1. Placement Cost Function

- Placement tries to minimize a cost function that captures multiple objectives like timing, wirelength, congestion and density. The total cost is a weighted sum of individual components.

$$\text{Total Cost} = w_1 \cdot \text{Timing Cost} + w_2 \cdot \text{Wirelength Cost} + w_3 \cdot \text{Congestion Cost} + w_4 \cdot \text{Density Cost}$$

where, w_1, w_2, w_3, w_4 → weights (tunable)

$$\sum w_i = 1$$

Components :

① **Timing Cost :** Penalizes setups / holds violations. Uses slack based or timing graph based models.
→ Higher violations → higher cost.

② **Wirelength Cost :** Estimated total interconnect length (HPWL). Lower wirelength → lower delay & power.

③ **Congestion Cost :** Penalizes overutilized routing resources (GCells / tiles / regions).
→ High congestion → high cost.

④ **Density Cost :** Maintains legal cell density in rows and regions. Avoids over / under density.
→ Density deviation → high cost.

Cost Optimization

- Global Placement : coarse optimization using analytical methods.
- Legalization : makes placement legal in rows.
- Detailed Placement : fine optimization (timing / congestion).

✱ **Key Point**

Placement engine continuously adjusts cell locations to minimize the cost function while satisfying placement constraints and design rules.

2. Types of Placements

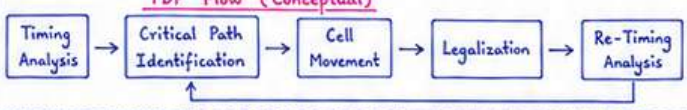
Placement strategies are chosen based on design goals and stage of implementation.

A. Timing Driven Placement (TDP)

- Main objective is to optimize timing (reduce negative slack).
- Moves cells to shorten critical paths and balance delays.
- Uses timing analysis, slack, fanout, criticality information.
- May increase congestion in some cases.
- Best for designs with tight timing requirements.

Focus : Improve Timing (WNS / TNS), Reduce Violations

TDP Flow (Conceptual)



Advantages
- Better timing closure.
- Reduces setup / hold violations.
- Improves WNS and TNS.

Limitations
- May create congestion.
- Not ideal in early stages.
- May increase wirelength.

B. Congestion Driven Placement (CDP)

- Main objective is to reduce congestion in routing resources.
- Distributes cells to balance routing demand.
- Uses congestion maps, global routing demand.
- May not produce best timing.
- Best for routability and early placement stages.

Focus : Improve Routability, Reduce Congestion

CDP Flow (Conceptual)



Advantages
- Low congestion.
- Better routability.
- Useful in early placement.

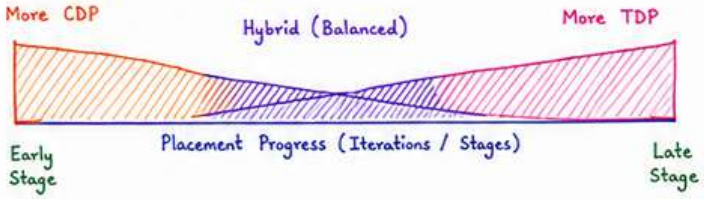
Limitations
- May worsen timing.
- Longer wirelength possible.
- Not ideal near sign-off.

3. TDP vs CDP Comparison

Parameter	Timing Driven Placement	Congestion Driven Placement
Primary Goal	Improve timing	Reduce congestion
Focus	Critical paths, slack	Routability, routing demand
Information Used	Timing analysis, slacks, criticality, delays	Global routing demand, congestion maps
Effect on Timing	Improves timing	May degrade timing
Effect on Congestion	May increase congestion	Reduces congestion
Wirelength	May increase	May increase slightly
Best Used	Mid to late stage	Early to mid stage
Result	Higher WNS, lower TNS	Lower congestion, better routability

4. Hybrid Placement Approach

- Modern tools use hybrid placement combining TDP and CDP.
- Cost function weights are adjusted dynamically.
- Early stages → Congestion focus.
- Later stages → Timing focus.



5. Practical Observations

→ Good placement is foundation for timing closure and routability.
→ Poor placement increases wirelength, congestion and power.
→ Always check WNS, TNS, congestion, density after placement.
→ Re-run placement iteratively with updated constraints.
→ Use region / blocks, grouping, and floorplan hints for better results.

6. Interview / Viva Points

Q. What is placement cost function ?
→ It is weighted sum of timing, wirelength, congestion and density costs.

Q. What is TDP ?
→ Timing Driven Placement focuses on reducing timing violations and improving slack.

Q. What is CDP ?
→ Congestion Driven Placement focuses on reducing routing congestion and improving routability.

7. Summary Points

✓ Placement minimizes cost function → timing + wirelength + congestion + density.
✓ TDP improves timing but may increase congestion.
✓ CDP reduces congestion but may worsen timing.
✓ Hybrid placement gives balanced results.
✓ Good placement is key for successful routing and timing closure.

8. Key Takeaway

Choose the right placement strategy based on design stage and priorities.

Early → CDP
Later → TDP
Best → Hybrid ✱

PLACEMENT

TYPES OF PLACEMENTS

1. AREA DRIVEN PLACEMENT (ADP)

- Main objective is to minimize total area.
- Compact placement with less whitespace.
- Improves area utilization.
- Fewer rows / steps → smaller core area.
- Good for area constrained designs.

Focus : Minimize Area, High Utilization

ADP Flow (Conceptual)

Global Placement → Density Control → Whitespace Removal → Legalization & Cleanup

Advantages

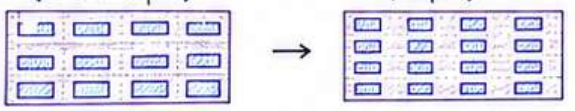
- Smaller die area.
- Better area utilization.
- Lower placement whitespace.
- Good for cost sensitive chips.

Limitations

- May increase congestion.
- Can hurt timing due to dense placement.
- Routability may degrade.
- More overlapping → timing risk.

Area Impact (Example)

Non-optimized placement (more whitespace) → Area driven placement (compact)



Result : Less whitespace → Smaller core area

2. POWER DRIVEN PLACEMENT (PDP)

- Main objective is to minimize total dynamic & leakage power.
- Places high switching activity cells near power sources.
- Balances power grid load.
- Reduces IR drop and switching power.
- Important for low power / battery operated designs.

Focus : Reduce Power, Balance Power Distribution

PDP Flow (Conceptual)

Switching Activity Analysis → Power Aware Cell Clustering → Power Balanced Placement → Legalization & Cleanup

Advantages

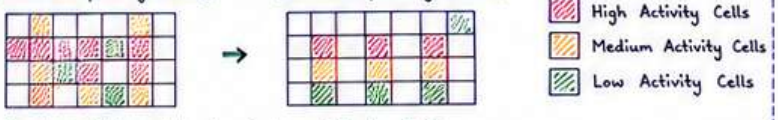
- Lower dynamic power.
- Less leakage power.
- Better power grid balance.
- Helps in IR drop reduction.

Limitations

- May increase wirelength.
- Can increase area & congestion.
- Timing may get degraded.
- Extra iterations needed.

Power Distribution Impact (Example)

Without Power Driven Placement (Unbalanced power grid load) → With Power Driven Placement (Balanced power grid load)



Result : Balanced load → Lower IR drop & Power

3. ROUTABILITY DRIVEN PLACEMENT (RDP)

- Main objective is to improve routability and reduce congestion.
- Distributes cells to balance routing demand.
- Reduces detours and via usage.
- Improves design manufacturability.
- Focused on reducing congestion hot-spots.

Focus : Improve Routability, Reduce Congestion

RDP Flow (Conceptual)

Global Placement → Congestion Estimation → Congestion Driven Optimization → Legalization & Cleanup → Congestion Re-Check

Advantages

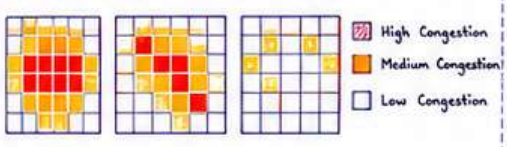
- Low congestion.
- Better routability.
- Fewer detours.
- Easier detailed routing.

Limitations

- May increase wirelength.
- Larger area.
- Slight timing impact.
- More iterations.

Congestion Map Impact (Example)

Before RDP (High Congestion) | During RDP (Optimization) | After RDP (Low Congestion)



Result : Balanced routing demand → Smooth routing

4. COMPARISON SUMMARY

Parameter	Area Driven Placement (ADP)	Power Driven Placement (PDP)	Routability Driven Placement (RDP)
Primary Goal	Minimize Area	Minimize Power	Improve Routability
Focus	High utilization, Less whitespace	Reduce dynamic & leakage power	Reduce congestion, Smooth routing
Key Input	Cell density, Area targets	Switching activity, Power grid	Routing demand, Congestion maps
Main Technique	Compact placement, whitespace removal	Power aware clustering, balancing	Congestion driven optimization
Effect on Area	Decreases area	May increase area	May increase area
Effect on Timing	May degrade timing	May degrade timing	Usually neutral / slight impact
Effect on Congestion	May increase congestion	May increase congestion	Reduces congestion
Effect on Power	Neutral / Slight increase	Significantly reduces power	Neutral
Best Used	Area constrained designs	Low power designs	Routing congested designs

5. PRACTICAL OBSERVATIONS

→ ADP gives smallest area but may increase congestion.

→ PDP reduces power but needs good power grid planning.

→ RDP is essential for smooth routing and timing closure.

→ Modern tools use hybrid approach combining all three.

6. INTERVIEW / VIVA Q&A

Q. What is Area Driven Placement ?

→ It minimizes whitespace and core area while meeting design constraints.

Q. What is Power Driven Placement ?

→ It places cells considering switching activity to reduce total power.

Q. What is Routability Driven Placement ?

→ It reduces congestion by balancing routing demand across the core.

7. SUMMARY POINTS

✓ ADP → Minimize area.

✓ PDP → Minimize power.

✓ RDP → Maximize routability.

✓ No single placement is best for all designs.

✓ Hybrid placement gives best overall results.

✓ Early placement quality decides final timing, power, area & routability.

8. KEY TAKEAWAY

Choose placement strategy based on design priority.

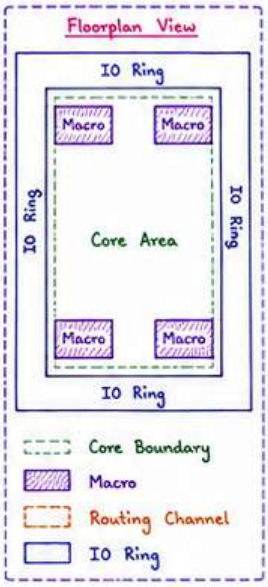
Area Priority → ADP
Power Priority → PDP
Routability Priority → RDP
Overall Best → Hybrid

PLACEMENT

PRE PLACEMENT CHECKLIST

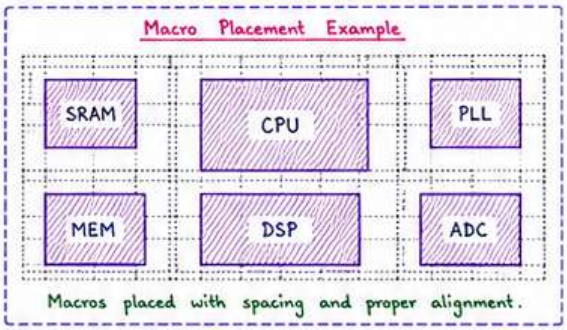
1. Floorplan Validation

- ✓ Die size, core area and aspect ratio as per spec.
- ✓ Core is properly defined and boundary is closed.
- ✓ Core area is sufficient for std. cells utilization.
- ✓ IO ring / Pad frame is complete and oriented.
- ✓ All IOs are placed on correct side with spacing.
- ✓ Keepout / Placement blockages are defined.
- ✓ Routing blockages (channels) are defined correctly.
- ✓ Well tap / End caps are enabled in std. cell rows.
- ✓ Site / Row height is correct as per technology.
- ✓ No std. cell rows outside core boundary.
- ✓ Scan chain stitch ports / test points provisioned.
- ✓ DFT, JTAG, spare cells regions reserved.
- ✓ Floorplan meets PPA targets (Area, Power, Timing).



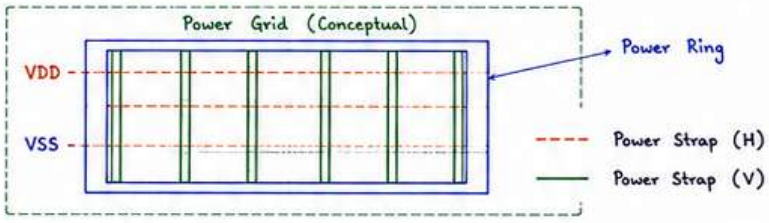
2. Macro Placement Checks

- ✓ All macros are placed within core.
- ✓ Macros are aligned to site / row grid.
- ✓ Sufficient spacing between macros maintained.
- ✓ Keepout around macros is defined.
- ✓ Macro orientation is correct.
- ✓ High fanout / critical macros placed centrally.
- ✓ Macro pins accessible for easy routing.
- ✓ No macros overlapping std. cell rows.
- ✓ No macros blocking major routing channels.
- ✓ Analog / Sensitive macros away from noisy blocks.
- ✓ Memory macros near power sources (if required).
- ✓ Macro halos / blockages are correctly applied.



3. Power Placement Checks

- ✓ Power ring (VDD/VSS) is complete and closed.
- ✓ Power ring width & spacing are as per EM/IR requirements.
- ✓ Power strap / stripes are defined in core.
- ✓ Strap pitch meets IR drop and EM criteria.
- ✓ Multiple power domains are correctly defined.
- ✓ Isolation cells / level shifters locations reserved.
- ✓ Decap cells planed region is reserved (if any).
- ✓ Power switch cells / headers are placed.
- ✓ Grid connectivity between rings and straps verified.
- ✓ PG nets (VDD/VSS) are continuous and no gaps.
- ✓ No std. cell rows on power ring / strap region.
- ✓ Power intent (UPF) matches with floorplan.



4. Utilization Checks

- ✓ Target utilization is within recommended range (60%–75%).
- ✓ Cell area / placeable area is calculated.
- ✓ Density uniformity across core is acceptable.
- ✓ No high density hot-spots in any region.
- ✓ Utilization per region (if defined) is within limits.
- ✓ Buffering demand (pre-CTS) is reasonable.
- ✓ Utilization after macro halo / blockages considered.
- ✓ Check whitespace availability for optimization.

Utilization Summary (Example)

Region	Cell Area (μm²)	Placeable Area (μm²)	Utilization (%)	Status
Core Overall	12.50 M	20.00 M	62.5 %	✓ OK
Region-1 (Left)	3.10 M	5.20 M	59.6 %	✓ OK
Region-2 (Center)	5.80 M	7.60 M	76.3 %	⚠ High
Region-3 (Right)	3.60 M	7.20 M	50.0 %	✓ OK

Target : 60% – 75% (Ideal ~ 65%)

Key Point

Pre-placement checklist ensures that the floorplan, macros, power and utilization are correct before placing standard cells. This avoids late stage rework, congestion and timing issues.

Practical Observations

- → Good floorplan + macro placement = Easy routing & timing closure.
- → Poor power planning = IR drop & reliability issues.
- → Unbalanced utilization = Congestion & timing degradation.
- → Always validate everything before starting global placement.

Interview / Viva Points

Q. Why pre-placement checklist is important ?
→ It ensures the design is ready for placement and avoids late corrections.

Q. What happens if power grid is not proper ?
→ High IR drop, timing failures and EM risks.

Q. What is ideal utilization range before placement ?
→ 60% – 75% (Ideal ~ 65%).

Q. Why macros need spacing and halo ?
→ For routing accessibility, power integrity and to meet DRC.

Summary Points

- ✓ Floorplan must be validated for area, aspect ratio and row correctness.
- ✓ Macros must be placed with spacing and good accessibility.
- ✓ Power grid must be strong and continuous.
- ✓ Utilization should be balanced and within target.
- ✓ Pre-placement checklist saves time and reduces iterations in later stages.

Quick Check Flow

Floorplan Validation
↓
Macro Placement Checks
↓
Power Placement Checks
↓
Utilization Checks
↓
Ready for Global Placement

PLACEMENT

PLACEMENT ALGORITHM - PARTITION BASED (MIN-CUT ALGORITHM)

1. DEFINITION

Placement algorithm decides the locations (X,Y) of all standard cells inside the core area such that total wirelength, timing violations, congestion and power are optimized while satisfying design constraints.

KEY GOAL OF PLACEMENT

- ✓ Minimize total wirelength
- ✓ Improve timing (reduce negative slack)
- ✓ Reduce congestion
- ✓ Meet power / IR drop goals
- ✓ Satisfy utilization and design rules

2. INTRODUCTION

- Placement converts abstract netlist into physical locations.
- It is the first major step after floorplan & power planning.
- Cells are moved from initial random locations to better locations using an objective (cost function).
- Partition based placement is scalable for large designs.
- Min-cut based partitioning balances wirelength & runtime.

3. WHY IT IS NECESSARY ?

- Random cell locations → long wires, high congestion.
- Good placement → better timing, power & routability.
- Provides legal starting point for CTS and routing.
- Reduces iterations in optimization and ECO.

✳ KEY POINT

Placement quality strongly impacts timing closure, power, area and routability of the final design.

4. PARTITION BASED PLACEMENT (MIN-CUT ALGORITHM)

A) INTRODUCTION

- Partition based placement divides the design into smaller partitions (blocks) and places them recursively.
- Min-cut algorithm partitions the netlist to minimize the number of nets (connections) cut between partitions.
- This reduces inter-partition wirelength and runtime.
- It uses recursive bisection / k-way partitioning.

BASIC IDEA

Cells connected together → keep in same partition
Nets cut between partitions → minimize (Min-Cut)

B) ALGORITHMS USED

- Min-Cut (Kernighan-Lin / Fiduccia-Mattheyses)
- Fast Greedy / Multi-level Min-Cut
- Recursive Bisection
- k-way Partitioning
- Coarsening & Refinement (Multi-level approach)

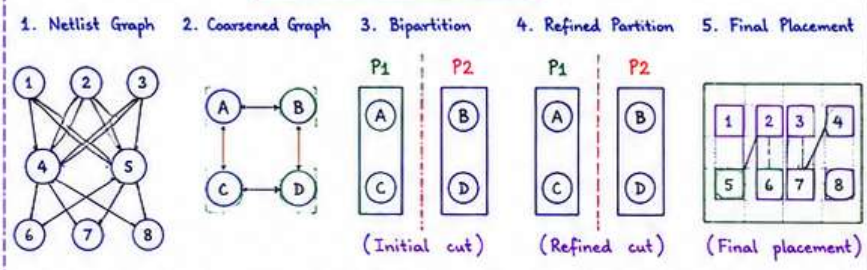
MIN-CUT

A cut is a set of nets whose pins are in different partitions.
Cutsize = Number of nets cut between partitions.
Goal: Minimize Cutsize while keeping partitions balanced.

C) STEP BY STEP PROCESS (MIN-CUT)

① Netlist	② Coarsening	③ Initial Partition	④ Refinement	⑤ Uncoarsening & Placement
Start with full netlist of cells & nets.	Merge cells into clusters to reduce problem size.	Partition the coarse graph into 2 or k parts.	Improve cut using FM or KL algorithm.	Expand clusters back and assign (x,y) locations.

PROCESS DIAGRAM (EXAMPLE)



D) STEP BY STEP DESCRIPTION

① **Build Netlist Graph**
Nodes → cells, Edges → nets.

② **Clustering / Coarsening**
Merge connected cells into clusters to reduce graph size.

③ **Initial Partition**
Partition the coarsened graph into two (bisection) or k parts.

④ **Refinement (Min-Cut Improvement)**
Apply FM / KL algorithm to swap cells and minimize cutsize.

⑤ **Uncoarsening**
Expand clusters back to individual cells level by level.

⑥ **Legalization**
Make cells legal in rows and remove overlaps.

⑦ **Final Placement**
Obtain (x,y) locations for all cells.

REFINEMENT IDEA (FM/KL)

- Try swapping a cell from one part to other.
- Calculate gain = (reduction in cut nets) − (increase in cut nets)
- Perform sequence of swaps with positive gain.
- Choose best prefix of swaps.

E) ADVANTAGES

- Works well for large scale designs.
- Reduces interconnect wirelength.
- Balances partition sizes.
- Multi-level approach gives fast runtime.
- Good starting point for optimizations.

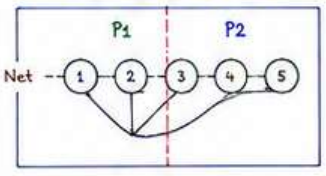
F) DISADVANTAGES

- Local minima possible.
- Quality depends on coarsening.
- May ignore timing / congestion initially.
- Not ideal for small blocks.
- Needs refinement & legalization.

G) APPLICATIONS

- Used in global placement of large ASIC / SoC.
- Used in hierarchical designs and blocks.
- Used before detailed placement and optimization.
- Used in EDA placement tools (Innovus, ICC2, Aprisa, etc.).

EXAMPLE (MIN-CUT)



Nets cut = 2
(Goal → Minimize)

H) PRACTICAL OBSERVATIONS

→ Good partitioning early → better timing & routing later.
→ Too many cuts → long wires.
→ Balanced partitions → better routability.
→ Always check congestion after global placement.

I). INTERVIEW / VIVA Q & A

Q. What is partition based placement ?
→ It partitions the design into smaller parts using min-cut and places them recursively.
Q. What is Min-Cut ?
→ It is the minimum number of nets cut between partitions.
Q. What algorithm is used for refinement ?
→ FM or KL algorithm.

J. SUMMARY POINTS

- ✓ Placement decides physical locations of cells.
- ✓ Partition based uses Min-Cut to reduce interconnect.
- ✓ Multi-level approach → coarsen → partition → refine → uncoarsen → place.
- ✓ Min-Cut minimizes number of nets cut.
- ✓ Balance partitions and reduce wirelength.
- ✓ Used for large designs before detailed placement.

K) QUICK RECAP FLOW

Netlist Graph
↓
Coarsening (Clustering)
↓
Min-Cut Partition
↓
Refinement (FM / KL)
↓
Uncoarsen & Place
↓
Legalization & Final Placement

PLACEMENT

PLACEMENT ALGORITHM

b) DIVIDE AND CONQUER ALGORITHM (D&C)

1. INTRODUCTION

- D&C algorithm recursively divides the placement area into smaller regions and places cells hierarchically.
- It reduces problem size at each level and combines solutions.
- It is one of the most widely used global placement techniques.
- Computationally efficient for large designs.

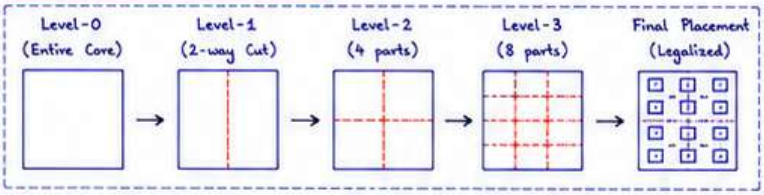
2. STEP BY STEP PROCESS

1. Start with entire core area
2. Partition area into k regions
3. Estimate cells / nets in each region
4. Recursively partition each region
5. Place cells inside smallest regions
6. Combine results & legalize placement

3. ALGORITHMS USED

- Recursive Bisection
- k-way Partitioning
- Sequence Pair
- FLUTE based clustering
- Bounding Box Partitioning

4. PROCESS DIAGRAM (EXAMPLE)



5. ADVANTAGES

- Scalable for large designs.
- Reduces runtime significantly.
- Hierarchical nature handles complexity well.
- Provides good initial placement.

6. DISADVANTAGES

- Early wrong cuts lead to poor final result.
- May ignore global connectivity.
- Not always optimal.
- Needs legalization & optimization.

7. APPLICATIONS

- Used in global placement of large ASIC / SoC.
- Suitable for hierarchical designs & blocks.
- Used in industry tools (ICC2, Innovus, Aprisa, etc.).
- Good for designs where partitioning is natural.

KEY POINT

* D&C algorithm partitions the design hierarchically, reduces problem size at each level and combines results to get efficient global placement.

c) QUADRATIC PLACEMENT ALGORITHM (QP)

1. INTRODUCTION

- Quadratic Placement formulates placement as a quadratic optimization problem.
- Cells are treated as objects connected by springs (nets).
- The goal is to minimize total wirelength (sum of squared net lengths).
- Solved using matrix based methods.

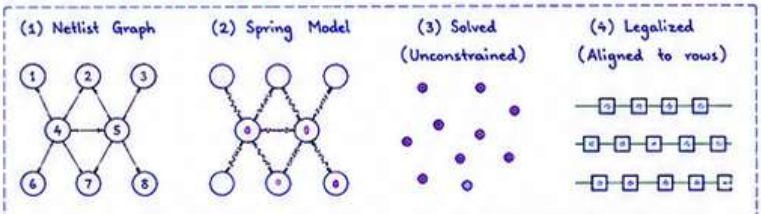
2. STEP BY STEP PROCESS

1. Build Netlist Graph
2. Compute Net Weights (based on fanout)
3. Formulate Quadratic Objective Function
4. Solve Linear System ($Ax = b$, $Ay = b$)
5. Obtain (x, y) locations for all cells
6. Legalization & Refinement (if needed)

3. ALGORITHMS USED

- Quadratic Programming (QP)
- Conjugate Gradient Method
- Cholesky Decompsition
- Simulated Annealing (post QP)
- Nesterov / Laplacian based QP

4. PROCESS DIAGRAM (EXAMPLE)



5. ADVANTAGES

- Considers global connectivity.
- Good wirelength optimization.
- Produces smooth, uniform distribution.
- Good starting point for detailed placement.

6. DISADVANTAGES

- Solving linear system is costly for very large designs.
- Produces unconstrained solution (needs legalization).
- Less effective on congestion.
- Memory usage can be high.

7. APPLICATIONS

- Used in early global placement stages.
- Used in small to medium size designs.
- Good for designs with uniform connectivity.
- Used in placement engines as optimization step.

KEY POINT

* QP models placement as a quadratic optimization problem and finds cell locations that minimize total wirelength while considering global connectivity.

D&C VS QUADRATIC PLACEMENT COMPARISON

Parameter	Divide and Conquer (D&C)	Quadratic Placement (QP)
Approach	Hierarchical partitioning	Mathematical optimization
Problem Model	Recursively cuts area & nests	Minimizes sum of squared wirelength (springs)
Scalability	Excellent for very large designs	Good for small to medium designs
Wirelength Quality	Good (depends on cuts)	Very good (global view)
Runtime	Fast	Moderate to Slow (solving linear system)
Congestion Handling	Limited	Better global distribution
Output	Discrete partitions	Continuous (needs legalization)
Use	Industry default for large designs	Used for optimization & academic tools

INTERVIEW / VIVA POINTS

Q. What is Divide and Conquer placement?
→ Hierarchically partitions area and places cells recursively.
Q. What is Quadratic Placement?
→ Solves a quadratic optimization problem to minimize total wirelength.
Q. Why D&C is preferred in industry?
→ Because it is fast and scalable for large designs.

SUMMARY POINTS

✓ D&C → Hierarchical partitioning based placement.
✓ QP → Optimization based global placement.
✓ D&C is faster and scalable.
✓ QP gives better wirelength optimization.
✓ Both produce initial placement for detailed stages.
✓ Legalization & refinement improve final results.

PRACTICAL OBSERVATIONS

→ Good partitions → better global placement.
→ Early bad cuts → timing & congestion issues.
→ QP needs good weights & proper solver.
→ Always legalize & refine after global placement.

KEY TAKEAWAY

Choose algorithm based on design size and goal.

Large designs → D&C
Better wirelength → QP
Balanced → Hybrid

PLACEMENT

PLACEMENT ALGORITHMS (Contd..)

d) FORCE DIRECTED PLACEMENT (FDP)

1. INTRODUCTION

- FDP models cells as objects in a physical system connected by springs (attractive) and repulsive forces.
- Attractive force for connected cells, repulsive force for all cells.
- Iteratively moves cells to minimum energy state.

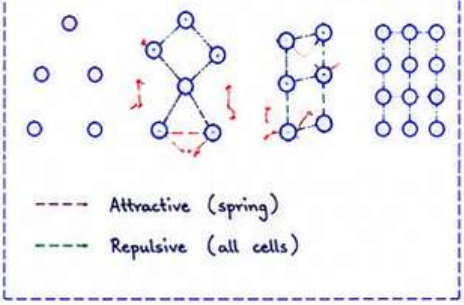
3. ALGORITHMS USED

- Spring Model (Hooke's Law)
- Coulomb Repulsion Model
- Gradual Step / Cooling Scheme
- Quadtree / Grid approximation for repulsion

2. STEP BY STEP PROCESS

1. Build Netlist Graph
2. Model Forces (attractive + repulsive)
3. Compute Net Forces on each cell
4. Move cells based on displacement vector
5. Update position & recompute forces
6. Repeat until convergence (stable placement)

4. APROCFENK DIAGRAM (EXAMPLE)



5. ADVANTAGES

- ✓ Produces good quality placement.
- ✓ Simple physical model, easy to understand.
- ✓ Can handle arbitrary floorplan shapes.
- ✓ Good starting point for refinement.

6. DISADVANTAGES

- ✗ Computationally expensive ($O(n^2)$).
- ✗ Convergence may be slow.
- ✗ Sensitive to initial conditions.
- ✗ Not suitable for very large designs without acceleration.

7. APPLICATIONS

→ Used in placement tools for global placement (especially for medium size designs).
→ Used as initial placement for timing driven / detailed placement.
→ Good for designs where connectivity is non-uniform.

✱ KEY POINT

Force Directed Placement = Cells as physical objects with forces.
Goal: Reach minimum energy state → good placement.

e) ANALYTICAL PLACEMENT (AP)

1. INTRODUCTION

- AP uses mathematical optimization to directly compute cell locations.
- Cells are treated as continuous variables and solved using linear algebra.
- Models wirelength using quadratic approximations.
- Very fast and scalable for large designs.

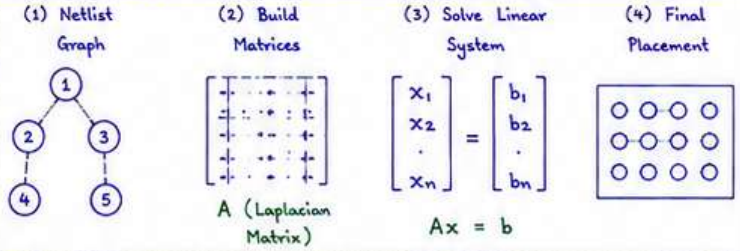
3. ALGORITHMS USED

- Quadratic Programming (QP)
- Conjugate Gradient (CG)
- Sparse Matrix Solvers
- Closed Form / Eigenvalue methods
- Laplacian Matrix based methods

2. STEP BY STEP PROCESS

1. Build Netlist Graph
2. Compute Net Weights (based on fanout / timing)
3. Formulate Objective Function (quadratic wirelength)
4. Solve Linear System ($Ax = b$)
5. Obtain (x, y) locations for all cells
6. Legalization & Refinement (if required)

4. PROCESS DIAGRAM (EXAMPLE)



5. ADVANTAGES

- ✓ Very fast and scalable (suitable for million + cells).
- ✓ Good global wirelength optimization.
- ✓ Deterministic and stable results.
- ✓ Provides excellent initial placement for refinement.

6. DISADVANTAGES

- ✗ Ignores discrete cell sizes initially.
- ✗ May overlap cells, requires legalization.
- ✗ Wirelength model is approximate.
- ✗ Not suitable as final placement alone.

7. APPLICATIONS

→ Used in global placement of large ASIC / SoC.
→ Used in industry standard tools (ICC2, Innovus, Aprisa, etc.).
→ Provides starting placement for timing driven optimization.

✱ KEY POINT

Analytical Placement = Mathematics based optimization.
Solves $Ax = b$ to get (x, y) locations → very fast & scalable.

COMPARISON OF FDP & AP

Parameter	Force Directed Placement (FDP)	Analytical Placement (AP)
Approach	Physics based (forces & springs)	Mathematical optimization (quadratic)
Model	Discrete iterative movement	Continuous optimization
Complexity	High ($O(n^2)$)	Low – Sparse Matrix ($O(n \log n)$ approx.)
Speed	Slow to Moderate	Very Fast
Scalability	Limited for very large designs	Excellent for million + cells
Quality	Good after convergence	Very good global wirelength
Overlaps	Less (natural spreading)	High (needs legalization)
Typical Use	Medium designs / Initial placement	Large designs / Global placement
Refinement Need	Moderate	High (legalization + timing driven)

INTERVIEW / VIVA Q & A

Q. What is Force Directed Placement ?
→ It uses attractive & repulsive forces to place cells iteratively to minimum energy.
Q. What is Analytical Placement ?
→ It uses mathematical optimization (QP) to directly compute cell locations.
Q. Which algorithm is faster ?
→ Analytical Placement is much faster.

SUMMARY POINTS

- ✓ FDP uses physical forces, good for medium designs.
- ✓ AP uses mathematical optimization, good for large designs.
- ✓ Both provide initial placement for refinement.
- ✓ Legalization & optimization improve final results.

PRACTICAL OBSERVATIONS

→ Poor initial placement = more iterations in later stages (CTS, Routing).
→ Analytical placement gives fast wirelength reduction.
→ FDP may give better spreading in non-uniform connectivity.

KEY TAKEAWAY

Choose placement algorithm based on design size, connectivity & goal.

Medium size → FDP
Large size → AP
Best results → Hybrid (AP + Refinement)

PLACEMENT

GLOBAL OPTIMIZATION METHODS & TIMING OPTIMIZATION IN PLACEMENT

1. GLOBAL OPTIMIZATION METHODS

- Global optimization is applied during placement to improve overall QoR in terms of timing, wirelength, congestion, power and area.
- It searches for better solutions beyond local minima by exploring large solution space.
- Global optimization techniques are used after initial placement and refinement iterations.

2. GOALS OF GLOBAL OPTIMIZATION

✓ Minimize total wirelength.
✓ Improve timing (reduce negative slack).
✓ Balance congestion and routing demand.
✓ Reduce power by lowering switching & clock load.
✓ Improve placement density distribution.
✓ Satisfy design rules and legalization.

3. COMMON GLOBAL OPTIMIZATION METHODS

① Mimlated Annealing (SA)
- Probabilistic search, escapes local minima.

② Tabu Search (TS)
- Uses memory to avoid cycling, explores better moves.

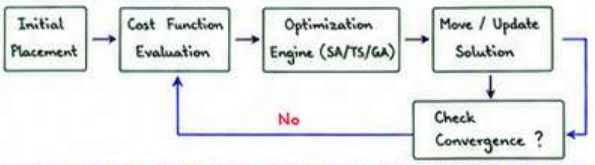
③ Genetic Algorithm (GA)
- Population based, uses crossover & mutation.

④ Mixed Size Optimization (MSO)
- Optimizes placement with cell sizing / buffering.

⑤ Particle Swarm Optimization (PSO)
- Swarm intelligence based search.

⑥ Machine Learning Guided Optimization
- Predicts good moves using ML models.

4. GLOBAL OPTIMIZATION FLOW



COST FUNCTION (EXAMPLE)

Total Cost = $\alpha \times$ Wirelength Cost + $\beta \times$ Timing Cost + $\gamma \times$ Congestion Cost + $\delta \times$ Power Cost

$\alpha, \beta, \gamma, \delta$ → weighting factors based on design priority.
Each term is calculated for current placement solution.

5. EXAMPLE - COST IMPROVEMENT

Iteration	Wirelength (μm)	Worst Slack (ns)	Congestion (%)	Total Cost
Initial	12.0 M	-0.35	72	1000
Iter - 10	9.8 M	-0.18	58	720
Iter - 20	8.1 M	-0.05	42	520
Iter - 30	7.2 M	+0.02	28	360 (Best)

6. ADVANTAGES

✓ Finds near optimal global solution.
✓ Improves overall timing and wirelength.
✓ Balances multiple conflicting objectives.
✓ Better QoR for large complex designs.

7. DISADVANTAGES

✗ High computational cost.
✗ Runtime increases with design size.
✗ Parameter tuning is critical.
✗ No guarantee of absolute optimum.

9. TIMING OPTIMIZATION IN PLACEMENT

1. INTRODUCTION

- Timing optimization in placement focuses on reducing negative slack and improving critical path delay.
- Cells on critical paths are placed closer to reduce data path delay.
- Requires timing driven cost function and analysis.

2. OBJECTIVES

✓ Minimize worst negative slack (WNS).
✓ Reduce total negative slack (TNS).
✓ Balance clock and data arrival times.
✓ Optimize setup & hold timing.

3. TIMING PARAMETERS

- Arrival Time (AT)
- Required Time (RT)
- Slack = RT - AT
- Setup Slack

  Setup Slack = RT - AT
- Hold Slack.

  Hold Slack = $AT - RT_{hold}$

4. TIMING DRIVEN PLACEMENT STRATEGIES

① Critical Path Weighting
- Increase weight of nets/cells on critical paths.

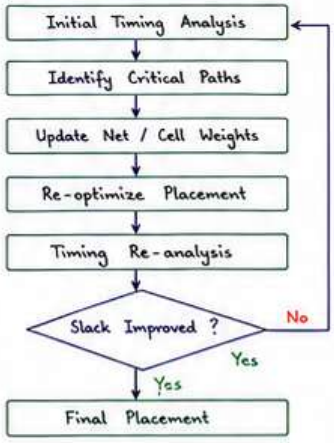
② Path Based Costing
- Consider full path timing during cost computation.

③ Timing Aware Net Weighting
- Weights updated based on slack values.

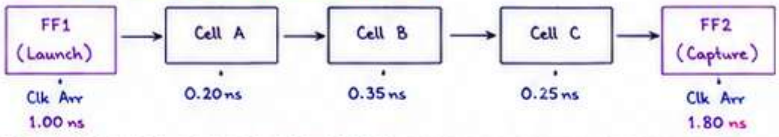
④ Optimization of Setup & Hold
- Fix setup violations first, then hold if needed.

⑤ Incremental Timing Updates
- Update timing only for affected paths.

5. TIMING OPTIMIZATION FLOW



6. EXAMPLE - CRITICAL PATH IMPROVEMENT



Initial Placement

Data Arrival = 1.00 + 0.20 + 0.35 + 0.25 = 1.80 ns
Required Time = 2.00 ns
Slack = 2.00 - 1.80 = +0.20 ns

After Timing Optimization

Data Arrival = 1.00 + 0.15 + 0.20 + 0.20 = 1.55 ns
Required Time = 2.00 ns
Slack = 2.00 - 1.55 = +0.45 ns (Improved)

7. ADVANTAGES

✓ Reduces negative slack.
✓ Improves critical path delay.
✓ Better timing closure.
✓ Enhances overall performance.

8. DISADVANTAGES

✗ May increase runtime.
✗ May worsen wirelength / congestion.
✗ Over optimization may cause hold violations.
✗ Needs accurate timing models.

9. PRACTICAL OBSERVATIONS

→ Early timing optimization in placement saves iterations later.
→ Balance timing with wirelength & congestion.
→ Use multi-objective cost function for best results.
→ Incremental optimization gives fast convergence.

✳ KEY POINT

Global optimization finds the best placement considering multiple objectives. Timing optimization in placement focuses on improving critical path slack by timing driven strategies.

INTERVIEW / VIVA Q & A

Q. What is global optimization in placement ?
→ It improves overall QoR by exploring large solution space using advanced methods.
Q. Why is timing optimization needed ?
→ To reduce negative slack and improve the critical path delay.
Q. Name some global optimization methods.
→ SA, TS, GA, PSO, MSO, ML based.

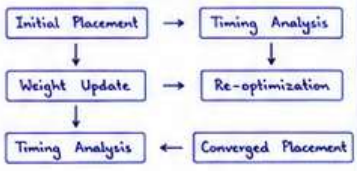
SUMMARY POINTS

✓ Global optimization improves overall QoR.
✓ Uses advanced search algorithms (SA, TS, GA, etc.).
✓ Timing optimization reduces WNS & TNS.
✓ Critical path weighting improves timing.
✓ Balanced cost function is essential.
✓ Leads to better timing, wirelength and congestion.

TIMING CONSIDERATIONS

- Setup: Data must arrive before clock edge.
- Hold: Data must be stable after clock edge.
- Slack > 0 → Good (GREEN)
- Slack < 0 → Bad (RED)
- Goal: Maximize slack.

QUICK RECAP FLOW



PLACEMENT

PLACEMENT DENSITY MAPPING

1. Introduction

- Density mapping is used to measure how much area in the core is occupied by standard cells.
- It helps in identifying over-utilized or under-utilized regions.
- Proper density distribution improves routability and timing closure.
- Density map is generated after global placement.

2. Why Density Mapping is Important ?

- High density regions → congestion & routing failures.
- Very low density regions → poor timing (long wires).
- Helps in cell spreading and legalization.
- Ensures uniform utilization across the core.
- Guides optimization iterations in placement flow.

3. What is Density ?

- Density is the ratio of the total cell area present in a region to the available placeable area of that region.
- It is usually expressed as a percentage.
- Calculated on a grid basis (e.g., 5μm x 5μm bins).

4. Density Formulae

$$\text{Cell Density (\%)} = \frac{\text{Total Cell Area in bin}}{\text{Placeable Area of bin}} \times 100$$

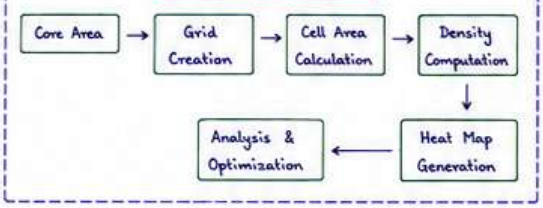
$$\text{Placeable Area} = \text{Bin Area} - \text{Blocked Area (Macros + I/O + Core Blockages)}$$

Typical Target Density Range : 60% - 75%
(Depends on technology & design rules)

5. Density Map Generation Steps

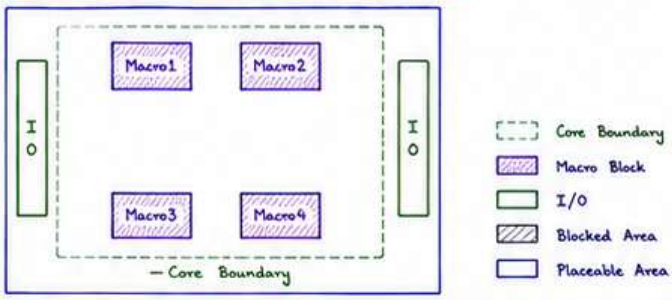
1. Divide the core area into small bins (grid).
2. For each bin, calculate total cell area present.
3. Subtract blocked area (macros, blockages, I/O).
4. Compute density using the formula.
5. Color-code the bins and generate density heat map.
6. Analyze and optimize if density is out of target range.

Density Map Flow

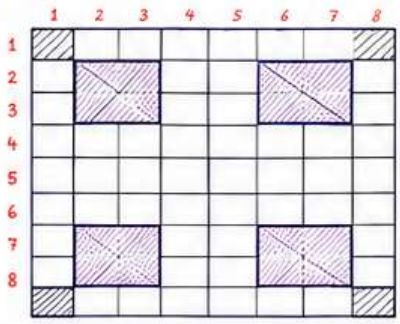


6. Density Map Example

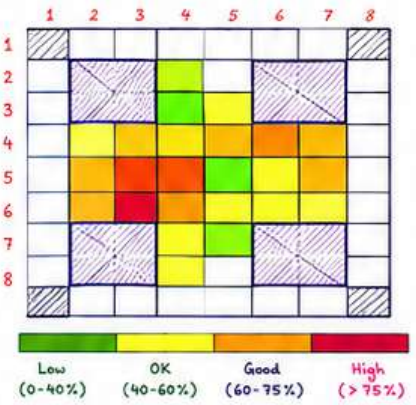
(A) Core with Macros (Floorplan View)



(B) Grid Division (Example 8x8 Bins)



(C) Density Heat Map (Color Coded)



(D) Density Table (Sample)

Bin (row, col)	Bin Area (μm²)	Blocked Area (μm²)	Cell Area (μm²)	Placeable Area (μm²)	Density (%)
(2,2)	100	0	42	100	42.0
(2,5)	100	0	68	100	68.0
(4,4)	100	60	20	40	50.0
(6,3)	100	0	80	100	80.0
(7,7)	100	0	30	100	30.0

(E) Example Observations

- Bins (2,5) & (6,3) → High density.
- Bin (7,7) → Low density.
- Bin (4,4) has macro blocking → low placeable area.
- Need to spread cells from high density bins to low density bins.

7. Density Classification

- Low Density : 0% - 40%
  → Under utilized, possible long wires, timing degradation.
- OK Density : 40% - 60%
  → Acceptable region.
- Good Density : 60% - 75%
  → Target / ideal range.
- High Density : > 75%
  → Congestion, routing failures.

8. Practical Observations

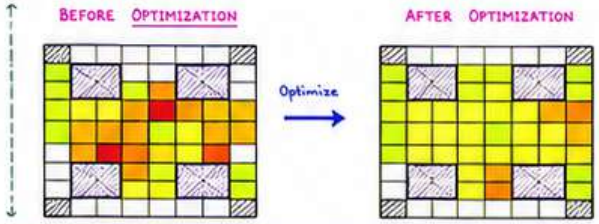
- Uniform density → better routability.
- Avoid large hot-spots (> 75%)
- Watch density near macros & blockages.
- Density maps are used iteratively during placement optimization.

* Key Point

Density mapping ensures uniform utilization of core area which leads to better routability, timing closure and overall design quality.

9. Using Density Map for Optimization

- If region density > target → move cells out.
- If region density < target → move cells in.
- Use legalization & cell spreading.
- Re-run placement & regenerate density map.
- Repeat until density is within target range.



10. Applications

- Global placement quality check.
- Congestion prediction.
- Utilization balancing.
- Timing improvement.
- ECO / Incremental placement.
- Chip finishing before CTS.

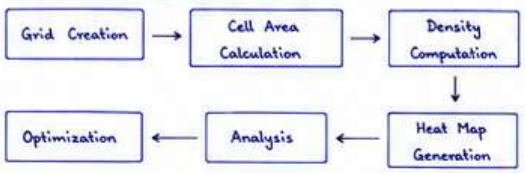
Interview / Viva Q & A

Q. What is placement density ?
→ Ratio of cell area to placeable area.
Q. Why density map is needed ?
→ To avoid congestion and long wires.
Q. What is ideal density range ?
→ 60% - 75% (depends on tech).
Q. What happens if density > 75% ?
→ Congestion and routing failures.

Summary Points

- ✓ Density = Cell Area / Placeable Area x 100.
- ✓ Generated on grid basis (bins).
- ✓ Ideal range is 60% - 75%.
- ✓ Helps in identifying congestion & idle regions.
- ✓ Used iteratively for placement optimization.
- ✓ Critical for timing, routability and PPA.

Quick Recap Flow



PLACEMENT

UTILIZATION AND CELL DISTRIBUTION ANALYSIS

1. INTRODUCTION

- Utilization & cell distribution analysis is performed after floorplan creation and before / during placement.
- It helps to understand how efficiently the core area is used and how cells are distributed across the core.
- Proper distribution avoids congestion, timing issues and ensures better routability.
- This analysis guides the placement engine to take better optimization decisions.

2. WHY UTILIZATION AND DISTRIBUTION ANALYSIS ?

- To check if the core area is sufficient for the design.
- To identify over-utilized (hot spots) and under-utilized regions.
- To balance cell distribution for better routing channels.
- To predict congestion and timing critical regions.
- To ensure uniform placement density across the core.

3. UTILIZATION

Utilization is the percentage of core area that is occupied by standard cells (placeable instances).

3.1 FORMULA

$$\text{Utilization (\%)} = \frac{\text{Total Cell Area}}{\text{Core Area} - \text{Blocked Area}} \times 100$$

Where,

- Total Cell Area = Sum of all standard cell areas.
- Core Area = Core Boundary Area.
- Blocked Area = Area occupied by macros, I/O, blockage, power structures (not available for placement).

3.2 IDEAL UTILIZATION RANGE

Technology Node	Recommended Utilization (%)	
	Typical	Maximum (avoid)
> 28nm	50% - 65%	< 70%
28nm - 7nm	55% - 70%	< 75%
≤ 5nm	60% - 75%	< 80%

3.3 EXAMPLE

- Core Area = 10 mm x 10 mm = 100 mm^2
- Blocked Area (Macros + IO + Blockages) = 20 mm^2
- Total Cell Area = 48 mm^2

$$\text{Utilization} = \frac{48}{(100 - 20)} \times 100 = 60\% \quad \checkmark \text{ (Good)}$$

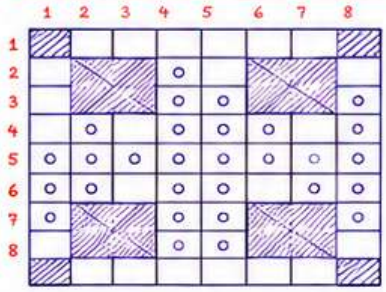
*** KEY POINT**

- Good utilization → better area efficiency.
- Too low utilization → more area, higher cost.
- Too high utilization → congestion, timing violations.
- Uniform distribution → fewer routing detours.

4. CELL DISTRIBUTION ANALYSIS

- Cell distribution shows how cells are spread across the core.
- It is analyzed using grids / bins or heat map.
- Helps to detect:
  - High density regions (hot spots).
  - Low density regions (under-utilized).
  - Imbalance across the core.
  - Potential congestion areas.

4.1 GRID BASED CELL DISTRIBUTION



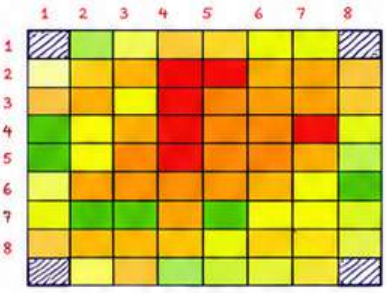
Blocked Area

O Cells in bin

Metrics from Grid

- Cells per bin
- Cell Area per bin
- Utilization per bin
- Overflow % (if > target)

4.2. CELL DISTRIBUTION HEAT MAP



Density Range (Utilization)

- 0% - 40% (Low)
- 40% - 60% (OK)
- 60% - 75% (High)
- > 75% (Critical)

Observation

- Center bins have high density.
- Corners are under-utilized.
- Need better cell spreading.

4.3 DISTRIBUTION CHECKLIST

- ✓ Density variation across core is within acceptable range.
- ✓ No bin exceeds maximum utilization limit.
- ✓ No large empty regions inside core.
- ✓ Cells are evenly distributed to leave routing channels.
- ✓ Congestion predictors are within target.

4.4. DISTRIBUTION FIXES

- → Re-balance placement constraints.
- → Adjust density and overflow targets.
- → Spread cells from high density bins to low density bins.
- → Use region based / cluster based placement.
- → Re-run placement optimization.

5. METRICS AND INTERPRETATION

Metric	Description	Good Range	Impact of Bad Value
Overall Utilization	% of core used by cells	50% - 75%	Too high → congestion Too low → more area
Avg Bin Utilization	Avg utilization of all bins	40% - 70%	High → congestion Low → area waste
Max Bin Utilization	Highest bin utilization	< 75%	> 75% → hot spot, routing failure
Std Dev of Bin Util.	Variation in bin utilization	< 15%	High variation → imbalance
Overflow %	% bins above target dens.	0% - 5%	High → congestion, timing issues

6. ADVANTAGES

- ✓ Ensures efficient area utilization.
- ✓ Identifies congestion & timing problem areas early.
- ✓ Improves routing success.
- ✓ Guides placement optimization.
- ✓ Better QoR (timing, power, area).

7. DISADVANTAGES

- × High analysis time for large designs.
- × Depends on floorplan quality.
- × Requires tuning of density targets.

8. APPLICATIONS

- Used in all placement flows.
- Early check before global placement.
- Iterative optimization during placement.
- Congestion driven placement decisions.
- Timing closure & routability improvement.

✱ PRACTICAL OBSERVATIONS

- → Poor utilization leads to higher cost & low efficiency.
- → Density hot spots create routing detours & timing violations.
- → Uniform distribution results in smooth routing.
- → Always check utilization & density after floorplan & before placement.

INTERVIEW / VIVA Q & A

Q. What is utilization ?
→ % of core area occupied by cells.

Q. Why is utilization important ?
→ To ensure efficient area usage and avoid congestion.

Q. What is cell distribution analysis ?
→ Analysis of how cells are spread across core.

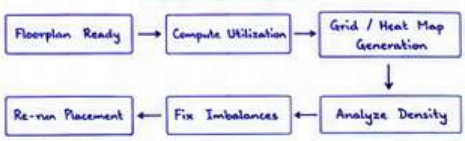
SUMMARY POINTS

- ✓ Utilization = Total Cell Area / (Core - Blocked) x 100
- ✓ Good utilization range = 50% - 75%.
- ✓ Distribution is checked using grid / heat map.
- ✓ Avoid high density hot spots (> 75%).
- ✓ Balanced distribution improves routability & timing.

DENSITY CLASSIFICATION

- Low Density : 0% - 40%
- OK Density : 40% - 60%
- High Density : 60% - 75%
- Critical Density : > 75%
- → Needs optimization

QUICK RECAP FLOW



PLACEMENT

DENSITY BASED OPTIMIZATION

1. INTRODUCTION

- Density based optimization balances cell concentration across the core.
- It reduces congestion, improves routability and helps timing closure.
- Density maps are used to guide cell distribution and spreading.
- Tools iteratively adjust placement to meet target density.

2. WHAT IS DENSITY ?

- Density shows how much cell area is present in a region / bin.
- High density → congestion, long wires.
- Low density → area under-utilization.

3. DENSITY DEFINITION & FORMULA

$$\text{Density (\%)} = \frac{\text{Cell Area in bin}}{\text{Bin Area}} \times 100$$

Where,

Cell Area in bin = Sum of standard cell areas inside the bin.

Bin Area = Physical area of the bin.

Important

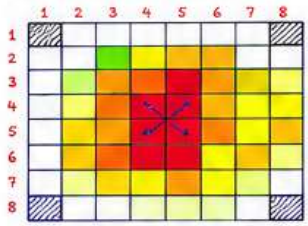
- Typical target density range : 60% - 80%
- Depends on technology, design rules & routing resources.

4. DENSITY MEASUREMENT PARAMETERS

Metric	Description
Cell Density	% of core area occupied by cells.
Overflow %	% bins above target density.
Underflow %	% bins below target density.
Local Density	Density in each bin.
Global Density	Overall utilization of core.

5. DENSITY MAP COMPARISON

(A) BEFORE OPTIMIZATION



(B) AFTER OPTIMIZATION





Low	OK	Good	High
(0-40%)	(40-60%)	(60-75%)	(>75%)

Observation Before Optimization :

- Hot spots in center & some regions.
- Overflow in many bins (>75%).
- Potential congestion & timing issues.

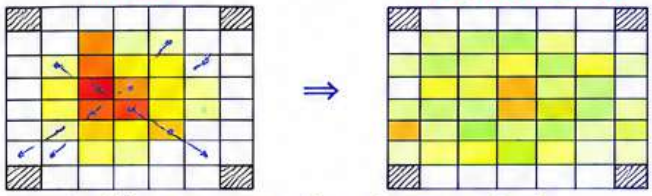
Observation After Optimization :

- Uniform density across core.
- Overflow reduced significantly.
- Better routability & timing closure.

7. HOW TOOL FIXES DENSITY PROBLEM ?

(A) CELL DISTRIBUTION (MOVE CELLS TOEEDED REGIONS)

BEFORE → AFTER DISTRIBUTION



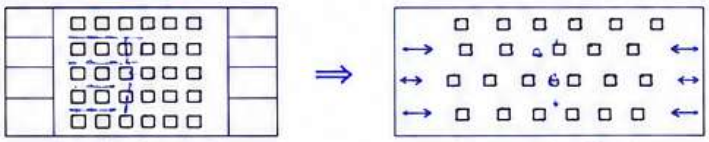
Tool identifies over-crowded bins and moves cells to under-utilized bins based on cost function.

Techniques Used :

- Cost based cell move.
- Bin based density targets.
- Gradient / quadratic cost models.

(B) CELL SPREADING (IMPROVE LOCAL DENSITY)

BEFORE → AFTER SPREADING



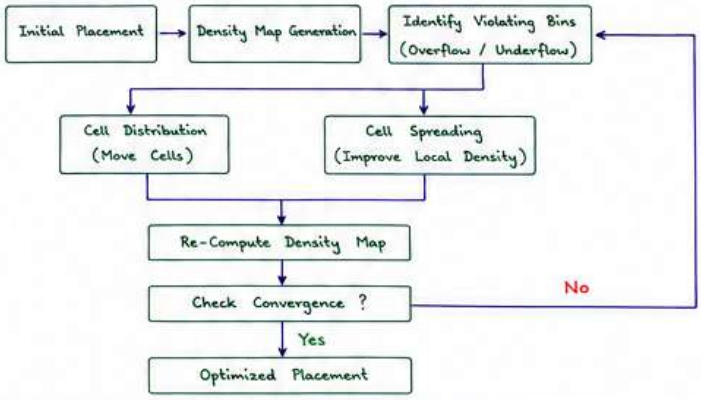
High density cluster — Cells spread uniformly

Tool spreads cells away from high density clusters to achieve target spacing.

Techniques Used :

- Cell spreading window.
- Minimize local density variance.
- Legalization & detour aware spreading.

8. DENSITY OPTIMIZATION FLOW



9. ADVANTAGES

- ✓ Reduces congestion.
- ✓ Improves routability.
- ✓ Better timing closure.
- ✓ Uniform power distribution.
- ✓ Efficient core area usage.

10. DISADVANTAGES / LIMITATIONS

- × Increased runtime.
- × May cause timing degradation if not controlled.
- × Over spreading increases wirelength in some cases.
- × Depends on accurate density targets.

INTERVIEW / VIVA Q & A

Q. What is density in placement ?
→ Ratio of cell area to bin area.

Q. Why density optimization is needed ?
→ To avoid congestion and improve routability & timing.

Q. What happens if density is high ?
→ Congestion, longer wires, timing problems.

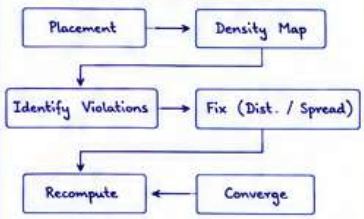
SUMMARY POINTS

- ✓ Density = Cell Area in bin / Bin Area x 100.
- ✓ Target density range : 60% - 80%.
- ✓ High density → overflow → congestion.
- ✓ Low density → underflow → area waste.
- ✓ Tool fixes density using distribution & spreading.
- ✓ Leads to better placement quality.

PRACTICAL OBSERVATIONS

- → Hot spots usually appear near macros / blockages.
- → Density optimization should be done iteratively with timing and congestion.
- → Always check density after macro placement and before CTS.

QUICK RECAP FLOW



PLACEMENT

DENSITY CONTROL MECHANISM

1. Introduction

- Even after global placement, some regions may have high cell density while others are under-utilized.
- Density control mechanism is used to balance cell density across the core.
- It improves routability, timing and overall QoR.
- Tools use spreading techniques to achieve uniform and timing aware distribution.

2. Definition

> Density control mechanism adjusts cell locations to maintain uniform density and remove hot spots while preserving timing and connectivity.

3. Why Density Control is Needed ?

- High density → congestion, routing detours, timing degradation.
- Low density → area waste, poor power efficiency.
- Balanced density → better routability & timing.
- Helps to meet design rules & target utilization.

4. Density Metrics

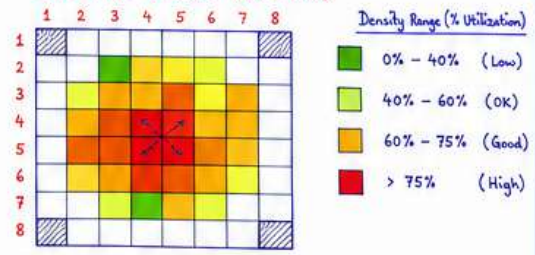
- Cell Density (%) = (Cell Area / Bin Area) × 100
- Overflow (%) = max(0, Density − Target Density)
- Underflow (%) = max(0, Target Density − Density)

Target Density Range

60% – 80% (depends on technology & design rules)

5. Density Map Example

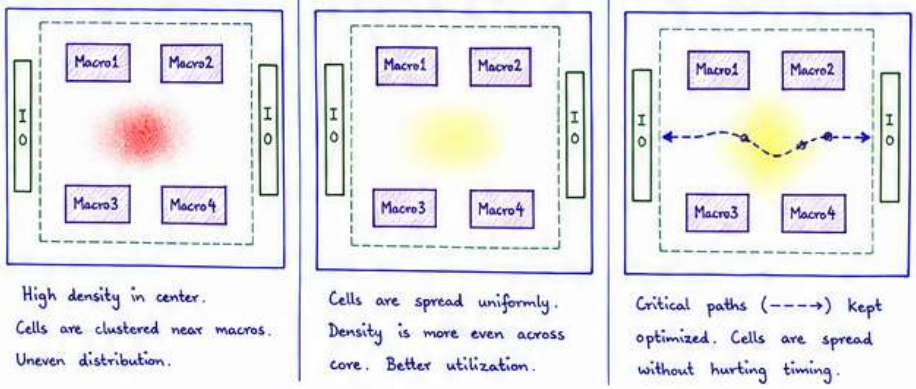
Grid Based Density (8 × 8 Bins)



Red region → Hot spot (high density)

Green region → Under-utilized (low density)

6. Comparison Images



(A) High density in center. Cells are clustered near macros. Uneven distribution.

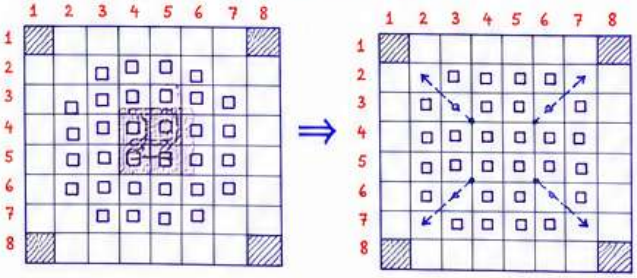
(B) Cells are spread uniformly. Density is more even across core. Better utilization.

(C) Critical paths (----→) kept optimized. Cells are spread without hurting timing.

----→ Core Boundary ▨ Macro Block ▭ I/O

7. How Tool Fixes Density Problem ?

(A) Cell Distribution (Move Cells Toeedd Regions)



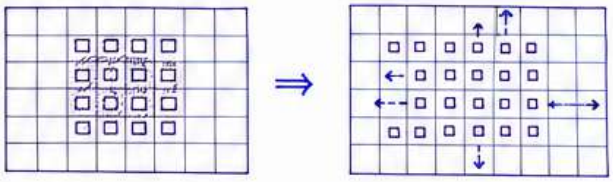
Techniques Used :

- Cost based cell move.
- Bin based targets.
- Gradient / quadratic cost models.
- Density driven wirelength optimization.

Effect :

- Reduces overflow bins.
- Improves routability.

(B) Cell Spreading (Improve Local Density)



Techniques Used :

- Cell spreading window.
- Penalty based spreading.
- Legalization with detour awareness.

Effect :

- Lowers local density peaks.
- Creates routing degradations.

(C) Timing Aware Spreading

▭ Standard Cell ● Critical Path Cells

8. Density Control Approaches Comparison

Approach	Objective	How it Works	Timing Impact	Use Case
Uniform Spreading	Uniform density across core	Move cells from bigs to low bins	Low	Improve routability & utilization
Local Spreading	Fix local hot spots	Spread cells within local window / neighborhood	Medium	Remove congestion in small regions.
Timing Aware Spreading	Maintain / improve timing	Spread while preserving critical paths	Low	Timing closure critical designs

9. Impact of Density Control

- ✓ Reconnects Reduced → better routing success.
- ✓ Timing Improved / Maintained → fewer violations.
- ✓ Wirelength Improved → lower interconnect delay.
- ✓ Power Improved → less switching due to detours.
- ✓ Area Utilization Balanced → no area waste.

*** Key Point**

Density control mechanism is critical after global placement to achieve uniform cell distribution, better routability and timing closure.

10. Advantages

- ✓ Improves routing resources.
- ✓ Reduces congestion & detours.
- ✓ Balances core utilization.
- ✓ Enhances timing & QoR.
- ✓ Helps in meeting target density.

11. Disadvantages / Limitations

- × Excessive spreading → longer wirelength.
- × May increase runtime.
- × Over spreading → timing degradation.
- × Depends on accurate models for critical designs.

12. Practical Observations

- → Always check density after global placement.
- → Keep density in 60% – 80% range.
- → Avoid large empty regions & hot spots.
- → Use timing aware spreading for critical designs.
- → Re-run placement & legalization after spreading.

13. Applications

- Congestion reduction in ASIC / SoC.
- Timing closure improvement.
- Power & area optimization.
- High performance & high density designs.

Interview / Viva Q & A

Q. What is density control mechanism ?
→ It balances cell density across the core.

Q. Why is density control important ?
→ To reduce congestion and improve routability & timing.

Q. What are density control techniques ?
→ Cell distribution, cell spreading, timing aware spreading.

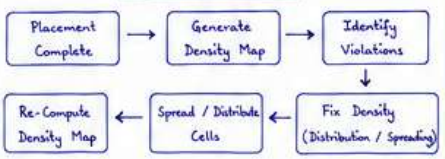
Summary Points

- ✓ Density control balances cell distribution.
- ✓ Identifies & fixes high density hot spots.
- ✓ Uses distribution & spreading techniques.
- ✓ Maintains target density (60% – 80%).
- ✓ Improves routability, timing & power.
- ✓ Critical for high performance designs.

Density Classification

- Low Density : 0% – 40%
- OK Density : 40% – 60%
- Good Density : 60% – 75%
- High Density : > 75% (Hot Spot)

Quick Recap Flow



PLACEMENT

CONGESTION ANALYSIS

1. Introduction

- Congestion analysis is performed after global placement to estimate routing difficulty.
- It compares routing demand with routing supply in each region of the chip.
- Helps to identify hot spots and guide optimization before detailed routing.
- High congestion may cause detours, timing violations and routing failures.

2. Why Congestion Happens ?

- Excessive cells packed in small area.
- High interconnect demand between modules.
- Poor floorplan or macro placement.
- Imbalanced placement of large macros.
- Insufficient routing resources (tracks / layers).
- Ignoring congestion during placement optimization.

3. Routing Demand

Routing demand is the total number of nets that want to pass through a routing resource.

- Demand is measured per routing resource (bin / track).
- Depends on number of nets and their span.
- Nets with longer span contribute more demand.

Demand Calculation (Simplified)

Demand (D) = $\sum$ (Net Weight × # of Bins / Tracks spanned by the net)

Where Net Weight = 1 (or based on criticality)

4. Routing Supply

Routing supply is the available routing capacity of a routing resource.

- Determined by routing layers, track pitch and width.
- Supply varies across the chip due to blockages, macros and power structures.

Supply Calculation (Simplified)

Supply (S) = # of Routing Tracks × Track Capacity per Track

(depends on layer, pitch, width, blockage density)

5. Routing Congestion

Congestion is the ratio of demand to supply in a routing resource.

Congestion (C) = Demand (D) / Supply (S)

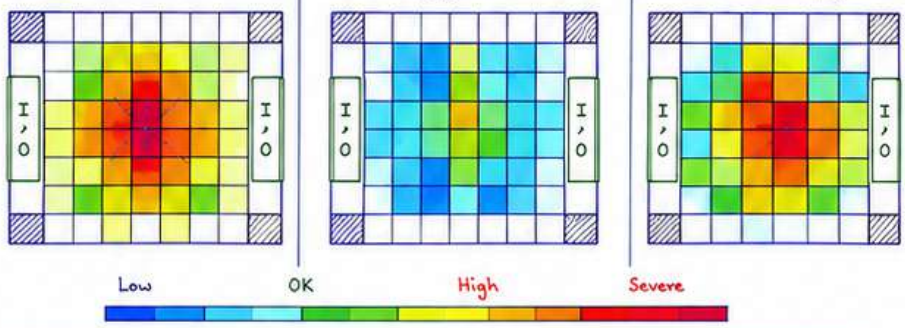
Congestion Ratio (C)	Severity	Interpretation
C ≤ 0.70	Low	Plenty of routing resources.
0.70 < C ≤ 1.00	Moderate	Acceptable, some risk.
1.00 < C ≤ 1.50	High	Difficult routing, timing impact.
C > 1.50	Severe	Very high risk of routing failure.

6. Visual Representation

(A) Routing Demand Map (More demand → Hotter color)

(B) Routing Supply Map (More supply → Cooler color)

(C) Congestion Map (Demand / Supply)

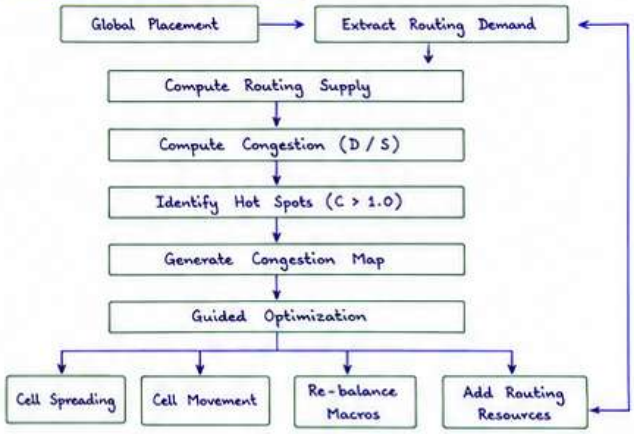


Low OK High Severe

Observations :

- Demand is high in center region.
- Supply is reduced near macros / blockages.
- Congestion is high in middle.
- Detours and longer wires may happen.

7. Congestion Analysis Flow



✳ **Key Point**

Congestion analysis helps to predict routing difficulty early and drives optimization for successful routing & timing closure.

8. Impact of Congestion

- Causes routing detours → longer wires → timing degradation.
- May require additional layers → area and cost increases.
- May lead to congestion failures during detailed routing.
- Affects IR drop, EM and signal integrity.

9. Practical Observations

- Always check congestion after global placement and before detailed placement.
- Hot spots often appear near macros, memories, IOs.
- Balanced floorplan and uniform placement reduce congestion.

10. Example – Region Wise Calculation

Region	Demand (D) (# of nets)	Supply (S) (# of tracks)	Congestion (C = D/S)	Status
R1 (Center)	1200	800	1.50	High
R2 (Top)	600	1000	0.60	Low
R3 (Left)	900	900	1.00	Moderate
R4 (Bottom)	750	1000	0.75	OK
R5 (Right)	1100	800	1.38	High

11. Remedies to Reduce Congestion

- ✓ Increase placement spreading.
- ✓ Move cells to under-utilized regions.
- ✓ Re-balance macro placement.
- ✓ Add / use more routing layers.
- ✓ Adjust floorplan to reduce blocking.
- ✓ Use congestion driven placement.
- ✓ Optimize net weights (critical nets).
- ✓ Early legalization & row alignment.

Interview / Viva Q & A

Q. What is congestion analysis in placement ?
A. It is the comparison of routing demand with routing supply to find difficult routing regions.
Q. What is a hot spot ?
A. A region where congestion ratio (D/S) > 1.0.
Q. How to reduce congestion ?
A. By spreading cells, moving macros, adding routing resources and optimizing placement.

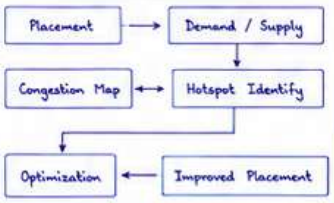
Summary Points

- ✓ Congestion analysis compares demand and supply.
- ✓ Demand increases with nets and their span.
- ✓ Supply depends on routing tracks and blockages.
- ✓ Congestion = Demand / Supply.
- ✓ High congestion → detours → timing issues.
- ✓ Hot spots identified by congestion maps.
- ✓ Optimization reduces congestion and improves QoR.

Congestion Classification

- Low : 0% – 70% (Good)
- Moderate : 70% – 100% (Acceptable)
- High : 100% – 150% (Difficult)
- Severe : > 150% (Routing Risk)

Quick Recap Flow



PLACEMENT

PIN DENSITY CONGESTION

1. Introduction

- Pin density congestion occurs when a specific region has too many cell pins packed closely.
- It limits routing resources and creates local routing bottlenecks.
- It is different from overall routing congestion (wire demand vs routing supply).

2. Definition

Pin Density is the number of cell pins within a given area (bin / window).

$$\text{Pin Density (PD)} = \frac{\text{Total Pins in Window}}{\text{Window Area}}$$

High pin density → more vias + tracks required
→ routing congestion & timing issues.

3. Why Pin Density Congestion Happens ?

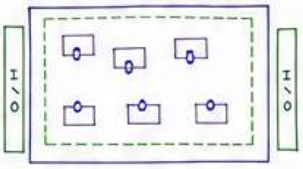
- Too many cells placed very close to each other.
- Small macros / blockages leave limited routing channels.
- High fanout nets concentrated in one region.
- Poor floorplan / tight channels.
- IO pins / macro pins near core edges.

4. Effects of High Pin Density

- More vias → via congestion.
- Tracks / routing resources get blocked.
- Increased detours → longer wires → timing degradation.
- Higher RC → more delay & power.
- Difficult detailed routing → routing failures.
- More DRC violations (min spacing / via spacing).

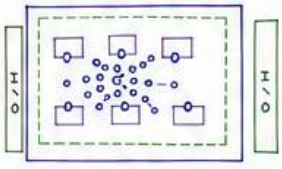
5. Examples

(A) Low Pin Density (Good)



- Pins are spread out.
- Enough space between cells.
- Easy routing with short wires.

(B) High Pin Density (Bad)



- Cells placed too close.
- Pins are clustered in center.
- Very less routing space.
- High via & track congestion.

(C) Pin Density Heat Map

0.3	0.5	0.8	1.4	1.2	0.4
0.4	0.7	1.1	1.8	0.9	0.5
0.6	1.0	1.6	2.3	1.1	0.7
0.5	0.9	1.4	2.0	1.7	0.6
0.4	0.6	1.0	1.5	1.2	0.4
0.3	0.5	0.7	1.1	0.9	0.3

Pin Density (pins / μm^2)

- 0 – 0.5 (Low)
- 0.5 – 1.0 (OK)
- 1.0 – 1.5 (Elevated)
- 1.5 – 2.0 (High)
- > 2.0 (Severe)

Interview / Viva Q & A

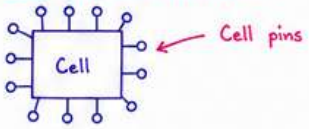
Q. What is pin density congestion ?
→ Too many cell pins in small area.
Q. Why it happens ?
→ Cells close together, fanout nets, small channels, macros, IOs.
Q. How to fix ?
→ Spreading, re-distribution, add tracks, add vias, multi-height cells.

Summary Points

- ✓ Pin density = Total Pins / Area.
- ✓ High pin density → via & track congestion.
- ✓ Causes detours, timing degradation & failures.
- ✓ Analyze using pin density heat map.
- ✓ Fix by cell distribution, pin spreading & adding routing resources.
- ✓ Check after placement & iterate.

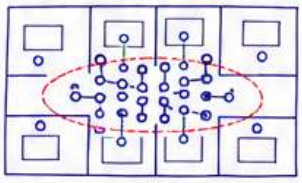
7. Pin Density Congestion Diagrams

(A) Cell Pin Representation



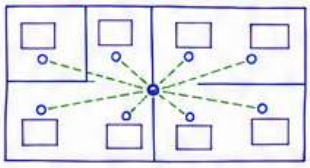
○ = Standard Cell Pin
● = Macro / IO Pin

(B) Clustered Pins (High PD)



Many pins in small area
→ via + track shortage
→ congestion.

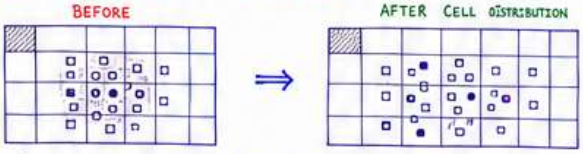
(C) Spread Pins (Low PD)



Pins are spread
→ routing space is available.

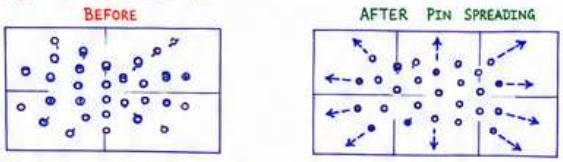
8. How to Fix Pin Density Congestion ?

(A) Cell Distribution / Spreading



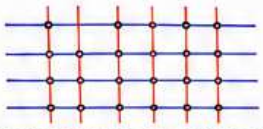
Cells are moved away from hot spot to distribute pins.

(B) Pin Spreading



Tool spreads pins using cell spreading or legalization.

(C) Add Routing Resources



- Add more routing tracks.
- Add extra vias.
- Increase routing layers.
- Use multi-height cells to reduce pins.

9. Practical Observations

- ✱ Check pin density after global placement and before CTS.
- ✱ Hot spots usually appear near macros, IOs and blockages.
- ✱ Balanced placement + spreading reduces pin density.
- ✱ High pin density regions lead to via stack congestion.
- ✱ Use pin density report and heat map from tool.

10. Advantages of Control

- ✓ Reduces via congestion.
- ✓ Improves routing success.
- ✓ Shorter wires & better timing.
- ✓ Less DRC violations.
- ✓ Better IR drop & EM performance.

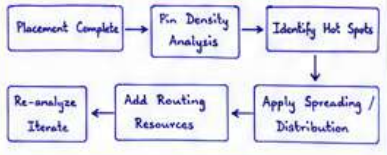
11. Disadvantages / Risks

- × Over spreading → long wires.
- × May increase if channels widened.
- × May break locality → timing impact.
- × Requires iteration & optimization.
- × Depends on accurate models.

Congestion Classification (PD Based)

- Low : 0 – 0.5 (Good)
- OK : 0.5 – 1.0 (Normal)
- Elevated : 1.0 – 1.5 (Watch)
- High : 1.5 – 2.0 (Critical)
- Severe : > 2.0 (Routing Risk)

Quick Recap Flow



PLACEMENT

MACRO CONGESTION

1. Introduction

- Macro congestion occurs when large macros obstruct routing resources and create uneven routing demand.
- It leads to high routing congestion, timing problems, IR drop and routing failures.
- Macro placement strategy greatly influences congestion.

2. Definition

Macro Congestion is the condition where macros / hard blocks cause excessive routing demand in nearby regions due to blockage or pin crowding.

It occurs when demand > available routing resources in a region.

3. Why Macro Congestion Happens ?

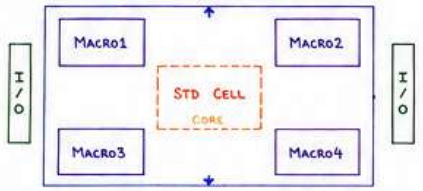
- Macros are large and block routing tracks / layers.
- High pin count macros concentrated in small area.
- Macros placed too close to each other.
- Macros near core edges / corners.
- Narrow channels between macros.
- Poor floorplan / unbalanced macro distribution.
- Ignoring routing resources while placing macros.

4. Effects of Macro Congestion

- High routing congestion → detours & longer wires.
- Timing degradation → setup / hold violations.
- Increased wirelength → more delay & power.
- Higher IR drop due to long power routes.
- Routing failures & DRC violations.
- Difficult detailed routing & more iterations.
- Area utilization becomes inefficient.

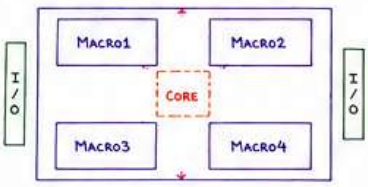
5. Examples

(A) Good Macro Placement (Low Congestion)



- ✓ Macros separated with good spacing.
- ✓ Wide channels for routing.
- ✓ Uniform routing demand.
- ✓ Low congestion.

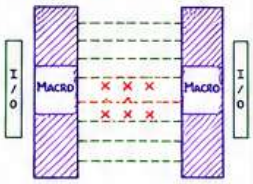
(B) Bad Macro Placement (High Congestion)



- ✗ Macros very close.
- ✗ Narrow / no channels.
- ✗ Pins concentrated.
- ✗ High routing demand.
- ✗ High congestion.

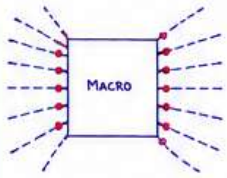
6. Macro Congestion Diagrams (Reason View)

(A) Blockage Effect



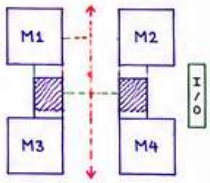
Macros block routing tracks in between. No direct paths.

(B) Pin Crowding Effect



High number of pins on one side → high local demand.

(C) Channel Narrowing



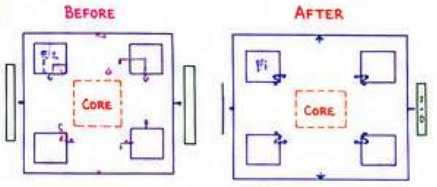
Very narrow channel between macros → congestion.

Legend: ▨ = Macro, ▭ = I/O, ● = Macro Pin, ---- = Routing Tracks, × = Blocked Tracks

7. How to Fix Macro Congestion ?

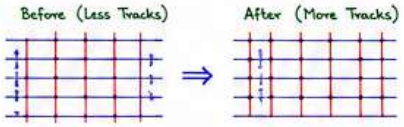
(A) Improve Macro Placement

- Distribute macros uniformly.
- Keep sufficient spacing.
- Avoid macros near core corners.
- Rotate / flip macros for better channels.



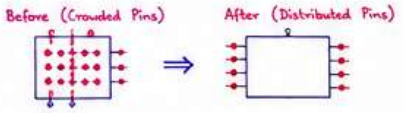
(B) Increase Routing Resources

- Increase routing layers.
- Increase track count.
- Use multi-height cells to reduce pin density.



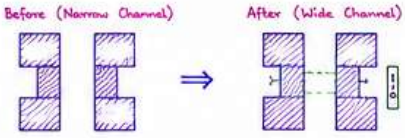
(C) Reduce Pin Crowding

- Distribute pins on multiple sides.
- Add pin access / feedthrough cells.
- Use hierarchical power / clocking.



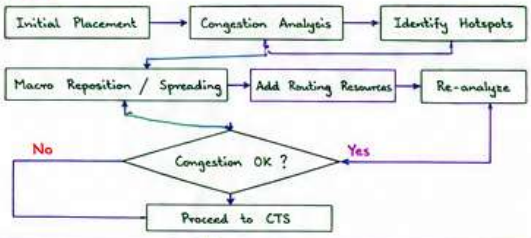
(D) Add / Widen Channels

- Reserve routing channels during floorplan.
- Increase channel width.
- Avoid blockage of channels.



(E) Routing Driven Placement

- Use congestion report during placement.
- Iterative placement + congestion fixing.
- Place macros using tool cost engines.



8. Macro Congestion Checklist

- ✓ Check congestion after global placement.
- ✓ Identify hotspots near macros.
- ✓ Verify pin density around macros.
- ✓ Ensure sufficient routing channels.
- ✓ Check macro spacing & distribution.
- ✓ Iterate placement until congestion is acceptable.

✱ Key Point

Macro congestion arises due to blockage, pin crowding and narrow channels created by macros. Good macro placement + sufficient routing resources + pin distribution are key to reducing congestion and achieving timing closure.

Impact Summary

Parameter	Without Fix	After Fix
Routing Congestion	High / Severe	Low / Acceptable
Wirelength	High	Reduced
Timing Slack	Negative	Improved
IR Drop	High	Reduced
Routing Success	Difficult	Easy

Practical Observation

- ✱ Always analyze congestion before CTS and routing.
- ✱ Macros near edges / corners create more congestion.
- ✱ Uniform macro distribution makes routing easier.
- ✱ Extra routing layers early in design saves iterations.

Interview / Viva Q & A

Q. What is macro congestion ?
→ Congestion caused by macro blockage and pin crowding leading to high routing demand.

Q. Why it happens ?
→ Large macros block routing resources, create narrow channels and pin crowding.

Q. How to fix it ?
→ Improve macro placement, add routing resources, spread pins and widen channels.

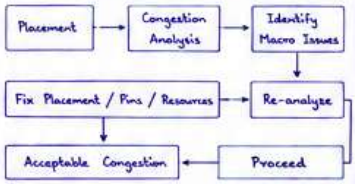
Summary Points

- ✓ Macro congestion limits routing resources.
- ✓ Caused by blockage, pin crowding & narrow channels.
- ✓ Leads to detours, timing issues & routing failures.
- ✓ Fixed by better macro placement, pin distribution, routing layers and channel widening.
- ✓ Iterative congestion analysis & optimization is essential.

Congestion Severity (PD Based)

- Low : 0% – 40% (Good)
- OK : 40% – 70% (Normal)
- Elevated : 70% – 100% (Watch)
- High : 100% – 150% (Critical)
- Severe : > 150% (Routing Risk)

Quick Recap Flow



PLACEMENT

CHANNEL CONGESTION

1. Introduction

- A channel is the routing space between two rows of standard cells or between cell rows and macros.
- Channel congestion occurs when routing demand in a channel exceeds available routing resources.
- It limits routability, increases detours and may cause timing and routing failures.

2. Definition

Channel Congestion is the condition where the total routing demand in a routing channel is more than the routing supply (capacity) of that channel.

Congestion (C) = Demand (D) / Supply (S)

C > 1 → Congested

3. Why Channel Congestion Happens ?

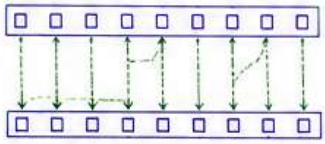
- Too many nets crossing the channel.
- High fanout nets concentrated in small region.
- Narrow channels due to large macros / blocks.
- Irregular cell placement leaving poor routing space.
- Ignoring routing resources while floorplanning.
- High pin count cells near channel edges.

4. Effects of Channel Congestion

- Routing detours and longer wirelength.
- Timing degradation due to extra delays.
- Increased congestion in adjacent layers / tracks.
- Hard to close timing and fix DRC.
- Possible routing failures (open nets).
- More power due to long routing & buffering.
- More iterations and design turn-around time.

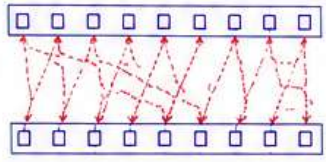
5. Examples

(A) Good Channel (Low Congestion)



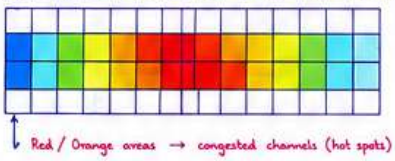
- ✓ Enough routing space
- ✓ Fewer crossing nets
- ✓ Easy for router
- ✓ C < 0.7 (Good)

(B) Bad Channel (High Congestion)



- × Many nets crossing
- × Routing tracks overloaded
- × Detours and blockages
- × C > 1.0 (Congested)

(C) Channel Congestion Heat Map (Demand / Supply)

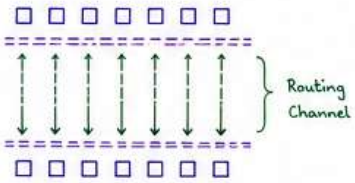


Red / Orange areas → congested channels (hot spots)

Congestion Range	
0 – 0.5	(Low)
0.5 – 1.0	(OK)
1.0 – 1.5	(High)
1.5 – 2.0	(Severe)
> 2.0	(Critical)

6. Channel Representation

(A) Standard Cell Rows

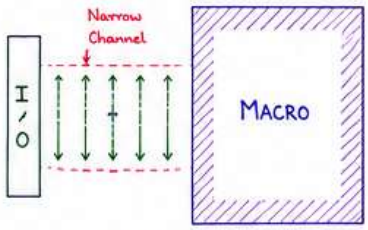


□ = Standard Cell

--- = Row Boundary

|||| = Available Tracks

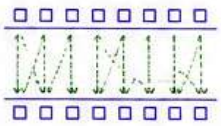
(B) Near Macro



Narrow channel between IO / macro causes high congestion.

7. Typical Channel Congestion Diagrams

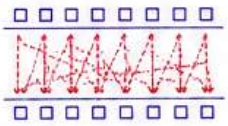
(A) Normal Channel



(B) Moderate Congestion



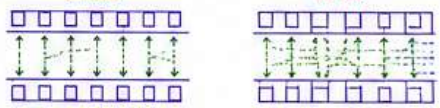
(C) High Congestion



Low Demand → Moderate → High Demand

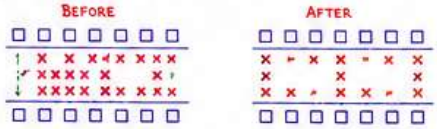
8. How to Fix Channel Congestion ?

(A) Widen Channels



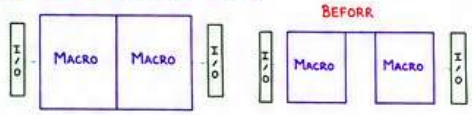
- Increase row spacing.
- Increase number of routing tracks.
- Improves routability.

(B) Balance Placement



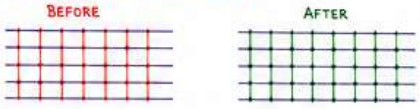
- Spread cells uniformly.
- Avoid clustering in one area.
- Reduces local demand.

(C) Move / Reposition Macros



- Move macros away from each other.
- Keep adequate spacing from core edges.
- Avoid narrow choke points.

(D) Add Routing Resources



- Add routing layers.
- Increase track count.
- Use multi-height cells.

(E) Routing Driven Placement



✱ **Key Point**

Adequate channel width and balanced placement are key to reduce congestion and achieve timing closure.

9. Interview / Viva Q A

Q. What is channel congestion ?

A. It occurs when routing demand in a channel is more than routing supply.

Q. What causes channel congestion ?

A. Too many nets, narrow channels, clustering, large macros, etc.

Q. How to fix channel congestion ?

A. Widen channels, balance placement, move macros, add routing resources.

Summary Points

- ✓ Channels are routing space between rows / macros.
- ✓ Congestion = Demand / Supply (C > 1 → bad).
- ✓ Caused by excess demand, narrow space & poor placement.
- ✓ Effects: detours, timing degradation, routing failures.
- ✓ Fixes: widen channels, spread cells, move macros, add routing layers, routing driven approach.

11. Congestion Severity (PD Based)

- Low : 0% – 70% (Good)
- OK : 70% – 100% (Normal)
- High : 100% – 150% (Critical)
- Severe : > 150% (Routing Risk)

Quick Recap Flow



PLACEMENT

EDGE CONGESTION

1. Introduction

- Edge congestion occurs near the die edges / core boundaries.
- It happens when routing demand close to the edge is more than the routing capacity available.
- It limits routability and may cause detours, timing violations and routing failures.

2. Definition

Edge Congestion is the condition where the routing demand in regions near die edges exceeds the routing supply (capacity) available in those regions.

Congestion (C) = Demand (D) / Supply (S)

C > 1 → Congested

3. Why Edge Congestion Happens ?

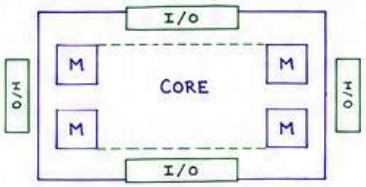
- High pin count cells / macros placed near edges.
- IO pins concentrated on one or more sides.
- Narrow channel width near edges (less routing tracks).
- Large macros blocking routing space near edges.
- Poor floorplan / core touching the die boundary.
- Ignoring edge routing resource while placement.

4. Effects of Edge Congestion

- Routing detours and longer wires.
- Timing degradation (more delay & skew).
- Increased IR drop and electromigration risk.
- Possible routing failures / open nets.
- Higher DRC violations (min spacing / via spacing).
- More iterations and design turn-around time.

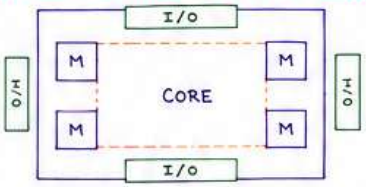
5. Examples

(A) Good Edge Placement (Low Congestion)



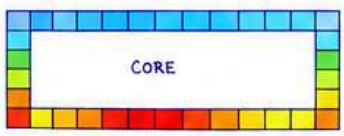
- ✓ Macros away from edges
- ✓ Sufficient space near edges for routing
- ✓ IOs distributed on multiple sides
- ✓ C < 0.8 (Good)

(B) Bad Edge Placement (High Congestion)



- × Macros near edges
- × IOs concentrated on one side
- × Narrow routing region near edges
- × C > 1.0 (Congested)

(C) Edge Congestion Heat Map (Demand / Supply)

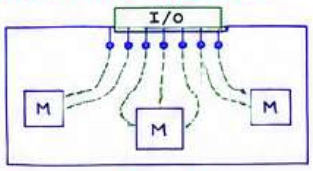


Red / Orange near edges → congested edges (hot spots)

Congestion Range	
0 - 0.5	(Low)
0.5 - 1.0	(OK)
1.0 - 1.5	(High)
1.5 - 2.0	(Severe)
> 2.0	(Critical)

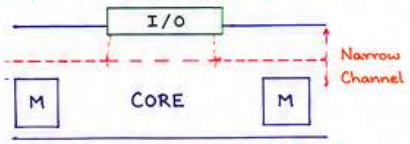
6. Edge Congestion Diagrams (Reason View)

(A) High IO Concentration



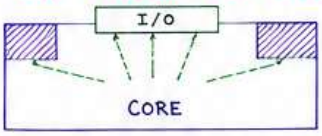
Too many IO pins on one side → high demand near that edge → congestion.

(B) Narrow Edge Channel



Less distance between Bndry & macros → less routing tracks.

(C) Macro Blockage Near Edge

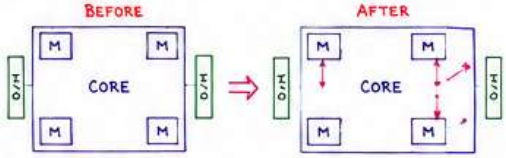


Macros near edge block tracks → detours & congestion.

Legend: = Macro, = IO IO Pins, = Pins, ---- = Routing Tracks, --- = Blockage / Narrow Channel

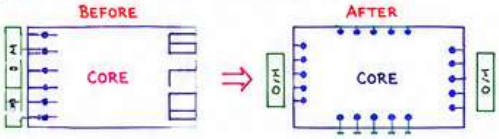
7. How to Fix Edge Congestion ?

(A) Move Macros Away From Edges



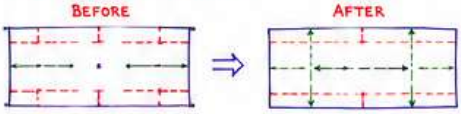
- Keep macros away from die edges.
- Add spacing for routing channels.

(B) Distribute IOs on Multiple Sides



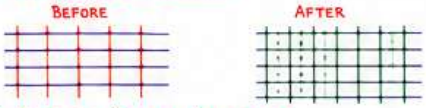
- Spread IOs across 2 or more sides.
- Reduce demand on single edge.

(C) Widen Edge Channels



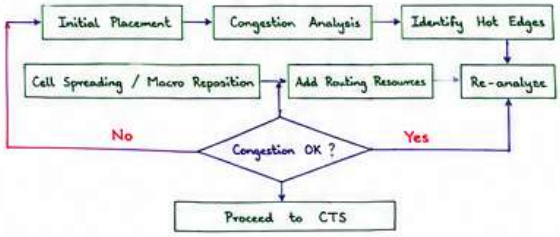
- Increase channel width near edges.
- Improves routing capacity.

(D) Add Routing Resources



- Add routing layers.
- Increase track count.
- Use multi-height cells.

(E) Routing Driven Placement



8. Interview / Viva Q & A

Q. What is edge congestion ?
→ More routing demand near die edges than available supply.
Q. Why does it happen ?
→ High IOs on one side, macros near edges, narrow channels.
Q. What are the effects ?
→ Detours, timing issues, multi-failures, IR drop, DRC violations.
Q. How to fix it ?
→ Move macros away, distribute IOs, widen channels, add tracks.

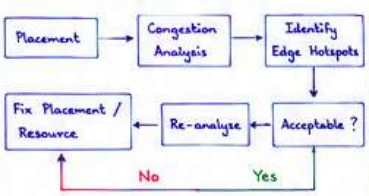
Summary Points

- ✓ Edge congestion occurs near die boundaries.
- ✓ Demand > Supply in edge channels.
- ✓ Causes detours, timing degradation and routing failures.
- ✓ Caused by high IO concentration, macros near edges, narrow channels.
- ✓ Fixes: macro reposition, IO distribution, widen channels, add routing resources.
- ✓ Check congestion after placement and iterate.

Congestion Severity (PD Based)

Low	0% - 40%	(Good)
OK	40% - 70%	(Normal)
Elevated	70% - 100%	(Watch)
High	100% - 150%	(Critical)
Severe	> 150%	(Routing Risk)

Quick Recap Flow



PLACEMENT

BLOCKAGES (HARD / SOFT / PARTIAL)

1. INTRODUCTION

- Blockages are regions in the core area where cell placement and/or routing is restricted.
- They are used to reserve space for macros, analog blocks, test structures, reserves and other constraints.
- Correct use of blockages helps in congestion reduction, timing closure and DRC compliance.

2. DEFINITION

A blockage is a user defined region that limits cell placement density and/or routing resources inside it.

Hard blockage → No cells or routes allowed.
Soft blockage → Cells allowed with limited density.
Partial blockage → Cells or routes allowed with partial capacity.

3. WHY BLOCKAGES ARE USED ?

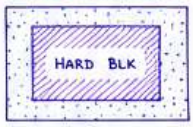
- Reserve space for hard macros / IPs.
- Protect analog / sensitive circuits from digital noise.
- Keep space for DFT, scan chains, clock structures.
- Reserve routing channels for high fanout nets.
- Meet technology / foundry keep-out rules.

4. EFFECTS OF BLOCKAGES

- Improper blockages → congestion & detours.
- Excessive hard blockages → area wastage.
- Too many soft blockages → placement density problems.
- Partial blockages misuse → routing capacity imbalance.
- Can help timing by guiding placement & routing.
- Reduce DRC, antenna & IR drop issues.

5. TYPES OF BLOCKAGES

(A) HARD BLOCKAGE



- ✓ No standard cells.
- ✓ No routing.
- ✓ Used for macros, analog IPs, memories, pads etc.

(B) SOFT BLOCKAGE (DENSITY BASED)



- ✓ Cells allowed with limited density.
- ✓ Routing allowed.
- ✓ Specified as max density %.
- ✓ Used to guide placement & avoid over crowding.

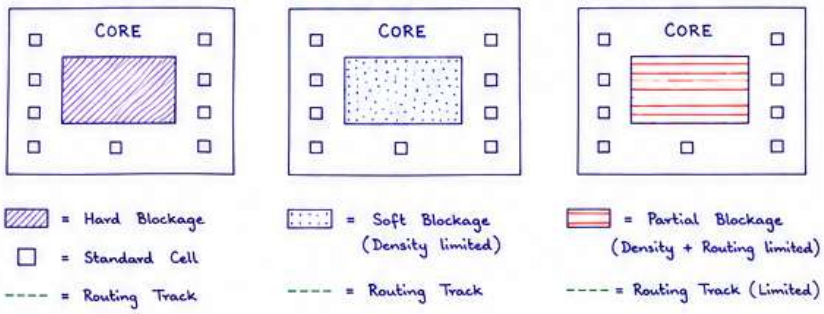
(C) PARTIAL BLOCKAGE



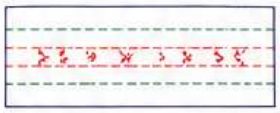
- ✓ Cells allowed with reduced density.
- ✓ Routing allowed with limited capacity.
- ✓ Specifies both cell density % and routing capacity %.
- ✓ Helps to balance placement & routing.

Type	Cell Placement	Routing	Typical Use
Hard Blockage	Not Allowed	Not Allowed	Macros, Analog IPs, Pads, Keep-outs
Soft Blockage	Allowed (Limited Density)	Allowed	Density control, Congestion relief
Partial Blockage	Allowed (Limited Density)	Allowed (Limited Capacity)	Channel reserve, Mixed constraints

6. BLOCKAGE DIAGRAMS (EXAMPLES)



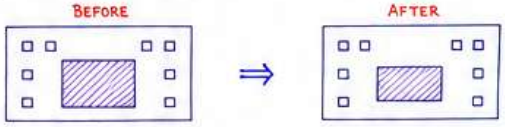
7. PARTIAL BLOCKAGE REPRESENTATION



Top & bottom → full routing capacity.
Middle region → partial routing capacity.
(Only limited number of tracks allowed)

8. HOW TO FIX BLOCKAGE RELATED ISSUES ?

(A) ADJUST BLOCKAGE SIZE



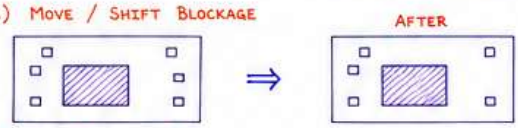
- ✱ Reduce over sized hard blockages.
- ✱ Release unnecessary reserved space.

(B) USE SOFT / PARTIAL INSTEAD OF HARD



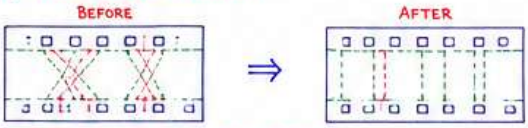
- ✱ Replace hard blockage with soft/partial.
- ✱ Allows placement with controlled density.

(C) MOVE / SHIFT BLOCKAGE



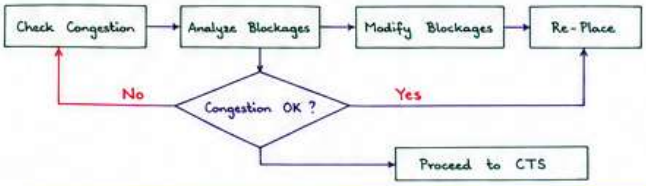
- ✱ Move blockage to less critical region.
- ✱ Keep channels open in congested area.

(D) ADD ROUTING CHANNELS



- ✱ Add dedicated routing channels between blockages.
- ✱ Increase spacing.

(E) GUIDED / RERUN PLACEMENT



9. PRACTICAL OBSERVATIONS

- ✱ Avoid huge hard blockages unless absolutely required.
- ✱ Always keep routing channels near large macros.
- ✱ Soft blockages are very useful for density control.
- ✱ Partial blockages help in channel reservation.
- ✱ Iterative placement with congestion analysis is important.

1. INTERVIEW / VIVA Q & A

Q. What is a blockage ?
→ A blockage is a region where placement and/or routing is restricted.

Q. Difference between hard and soft blockage ?
→ Hard : No cells, no routes.
Soft : Cells with limited density, routes allowed.

Q. Why partial blockage is used ?
→ To limit both cell density and routing capacity in a region to balance resources.

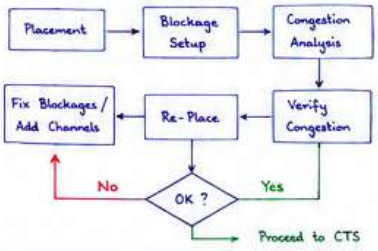
SUMMARY POINTS

- ✓ Blockages control placement density and routing in a region.
- ✓ Hard blockages completely restrict cells and routing.
- ✓ Soft blockages allow cells with limited density
- ✓ Partial blockages limit both cells and routing.
- ✓ Proper blockages reduce congestion and improve timing and DRC.
- ✓ Wrong blockage usage leads to area/timing issues.

10. CONGESTION SEVERITY (PD BASED)

- Low : 0% - 40% (Good)
- OK : 40% - 70% (Normal)
- Elevated : 70% - 100% (Watch)
- High : 100% - 150% (Critical)
- Severe : > 150% (Routing Risk)

QUICK RECAP FLOW



PLACEMENT

BLOCKAGE TYPE - FENCE

1. Definition

- Fence (Fence Region) is a soft type of blockage used in placement to control and guide the movement of standard cells / macros.
- It allows cells to move within the fence boundary, but restricts them from going outside.

> Fence does not block placement completely.
> It creates a preferred region for cells.

2. Characteristics

- ✓ Soft blockage (density based).
- ✓ Cells are allowed to move inside the fence.
- ✓ Cells outside the fence will avoid entering it unless floorplanning / constraints force them.
- ✓ Helps in area partitioning and resource reservation.
- ✓ Improves placement quality and reduces congestion.
- ✓ Fence can be applied to standard cells and macros.

3. Why Fences are used ?

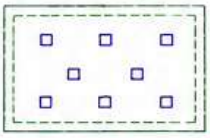
- Guide placement to desired regions.
- Keep critical blocks / logic near specific macros / IPs.
- Avoid interference between sensitive blocks.
- Reserve area for future ECO, decap, CTS, routing channels.
- Improve timing by grouping related cells.
- Control cell spreading and maintain locality.

4. Effects / Benefits

- Better placement control and predictability.
- Reduced congestion and routing detours.
- Improved timing and shorter critical paths.
- Helps in hierarchical / block based design flows.
- Improved routuity and imied IR drop performance.
- Supports ECO by reserving space.

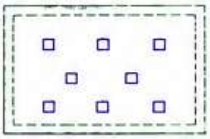
5. Fence Types (Sub-Types)

(A) Hard Fence (Strict Fence)



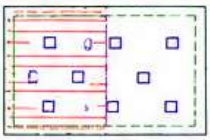
- ✓ Cells cannot cross the fence boundary.
- ✓ Strict region control.
- ✓ Used for hard partitioning of blocks.

(B) Soft Fence (Density Fence)



- ✓ Cells preferred inside the fence region.
- ✓ Density penalty if cells go outside.
- ✓ Most commonly used.

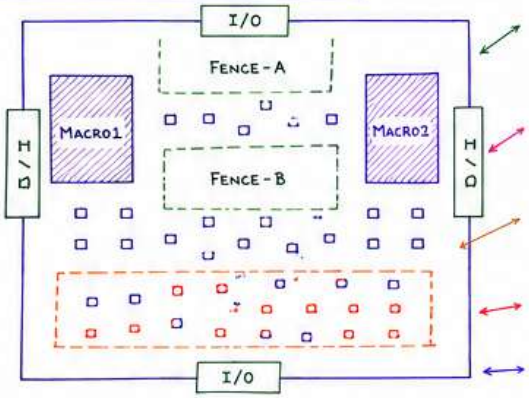
(C) Partial Fence (Mixed Constraint)



- ✓ Only part of the area is fenced.
- ✓ Used when only a portion needs reservation.
- ✓ Flexible control.

LEGEND: ⌜⌟ := Standard Cell; = Macro / IP; = Fence Boundary; = Soft Density Influence; = Partial Fence Area; |||| = Routing Channel

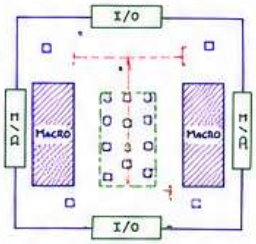
6. Fence - Diagram Explanation



- FENCE-A : Cells are preferred to stay inside this top region.
- FENCE-B : Reserved region for logic group near MACROs.
- FENCE-C : Area reserved for ECO / decap.
- Red dashed lines : Routing Channels.
- Cells outside fence will avoid entering unless necessary.

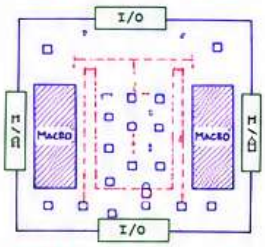
7. Fence Examples

(A) Good Fence Usage



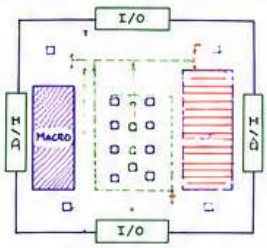
- Fences guide cells near macros.
- Less congestion, short wires.
- Good timing and routability.

(B) Poor / No Fence



- Cells scattered everywhere.
- More congestion in channels.
- Long wires, timing issues.

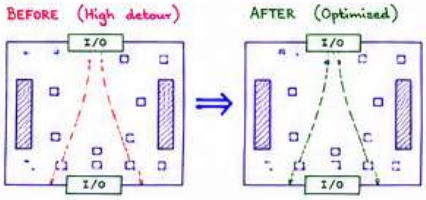
(C) Partial Fence



- Only part of region is fenced.
- Used for space reservation.
- Good flexibility.

8. How to Fix Fence Related Issues ?

- Adjust fence size / boundary.
- Modify density penalty.
- Move or resize macros.
- Add routing channels near fences.
- Use partial fence instead of hard fence.
- Perform congestion analysis and iterate.
- Re-balance placement after fence tuning.



9. Applications of Fence

- Place logic near specific macros / IPs.
- Reserve area for future ECO.
- Keep high speed logic in small region.
- Create regions for power islands.
- Reserve space for decap / filler cells.
- Avoid cells in analog / sensitive areas.
- Partition design for hierarchical flow.

10. Fence vs Other Blockages

Type	Blocks Placement ?	Allows Cells Inside ?	Density Control	Typical Use
Hard Blockage	Yes (No)	No	Not applicable	Macros, Pads, Keep-out, DRC
Soft Blockage	No	Yes (Density limited)	Yes	Density control, Congestion relief
Partial Blockage	No (Partially)	Yes (Limited)	Yes (Partial)	Space reservation, ECO, Channels
Fence (Soft)	No (Soft)	Yes (Preferred)	Yes (Penalty based)	Placement guide, Partitioning

✱ **Key Point**

Fence is a soft blockage used to guide placement and control cell movement within a preferred region without completely blocking the area.
It improves placement quality, timing and routability.

Quick Recap Flow

Define Fence Region → Apply Density Penalty → Run Placement → Analyze Congestion → Adjust Fence / Density → Re-Place & Optimize

11. Interview / Viva Q & A

Q. What is fence blockage ?
Soft region used to guide placement.

Q. Difference between hard blockage and fence ?
→ Hard blocks completely, fence allows placement inside.

Q. Why fences are used ?
→ Guide cells, reserve area, improve routing & timing.

Summary Points

- ✓ Fence is soft blockage that guides placement.
- ✓ Cells preferred inside fence region.
- ✓ Improves timing and routing.
- ✓ Used for area partitioning and reservation.
- ✓ Soft penalty keeps cells inside region.
- ✓ Essential for macro based and SoC designs.

12. Congestion Severity (PD Based)

- Low : 0% - 40% (Good)
- OK : 40% - 70% (Normal)
- Elevated : 70% - 100% (Watch)
- High : 100% - 150% (Critical)
- Severe : > 150% (Routing Risk)

Blockage Types Summary

- ✱ Hard → No placement allowed.
- ✱ Soft → Placement allowed with density limit.
- ✱ Partial → Partial region reserved.
- ✱ Fence → Placement guided, cells preferred inside.

PLACEMENT

BLOCKAGE TYPE - REGION (PLACEMENT BOUNDS)

1. DEFINITION

- Region Bounds (Placement Bounds) is a constraint that defines an area where cells are allowed to be placed.
- It is used to control cell density distribution and guide placement to desired regions.
- It does not block placement completely. It only allows cells inside the defined boundary.

> Region bounds help to balance density, reduce congestion and maintain design intent while keeping flexibility.

2. WHY REGION BOUNDS ARE USED ?

- Control cell distribution and density.
- Avoid congestion in critical regions.
- Reserve space for macros, IPs, analog, memories, ECO / decap, routing channels.
- Improve timing by keeping related cells closer.
- Partition design hierarchically and maintain floorplan intent.
- Useful for multi-power, multi-voltage designs.

3. CHARACTERISTICS

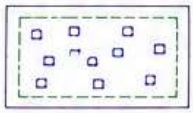
- ✓ Soft constraint (density based).
- ✓ Cells inside the region obey density rules.
- ✓ Cells outside region are discouraged / limited.
- ✓ Can have maximum density % and minimum density %.
- ✓ Can be rectangular, polygonal or any user defined shape.
- ✓ Overlapping regions possible with different priorities.
- ✓ Works at global placement and optimization stages.

4. EFFECTS / BENEFITS

- Better control over placement density and distribution.
- Reduced congestion and routing detours.
- Helps timing by keeping critical cells in defined area.
- Reserves space for future ECO, decap, fillers.
- Improves routability and IR drop performance.
- Maintains design floorplan intent.

5. TYPES OF REGION BOUNDS

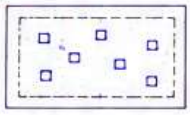
(A) HARD REGION BOUND (STRICT)



Inside ✓
Outside ✗

- ✓ Cells allowed only inside.
- ✓ No cells outside allowed.
- ✓ Strict density control.
- ✓ Used for critical blocks.

(B) SOFT REGION BOUND (DENSITY BASED)



Inside ✓
Outside (Limit)

- ✓ Cells preferred inside.
- ✓ Outside allowed with density penalty.
- ✓ Most commonly used.

(C) PARTIAL REGION BOUND



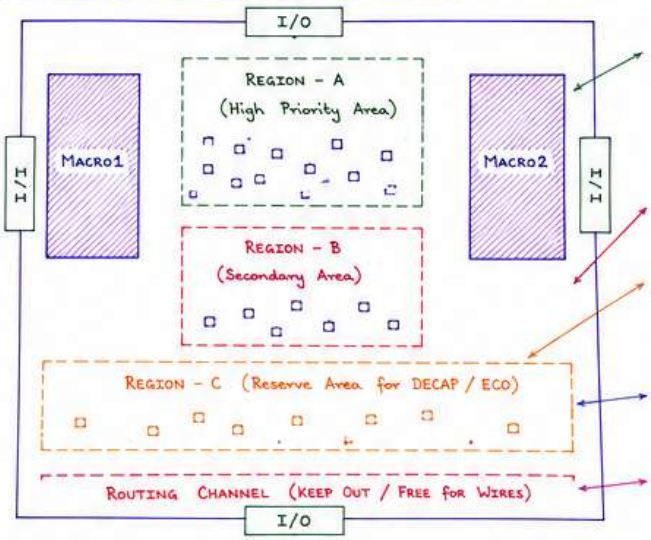
Inside (Priority)
Outside (Secondary)

- ✓ Only part of chip is region.
- ✓ Priority to inside.
- ✓ Outside allowed if space needed.

LEGEND :

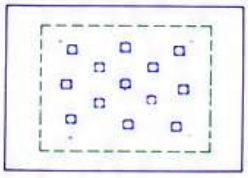
- = Standard Cell
- = Region Bound (Soft / Hard)
- = Cells Outside Region
- = Macro / IP
- = Partial Region
- = Routing Channel

6. REGION BOUNDS - DIAGRAM EXPLANATION



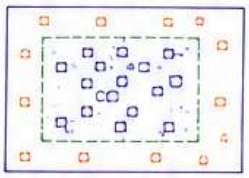
7. REGION BOUNDS - EXAMPLES

(A) HARD REGION BOUND



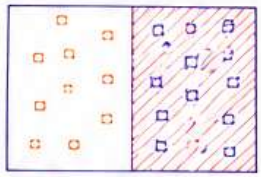
- Strict boundary.
- No cells outside.
- Common for critical IP or analog area.

(B) SOFT REGION BOUND



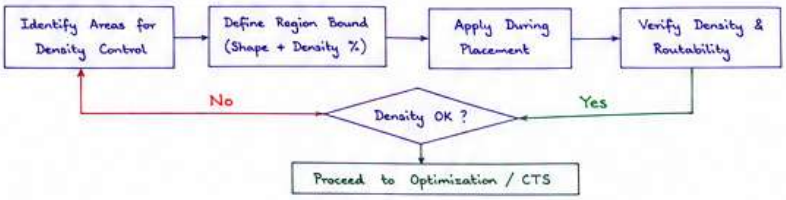
- Soft density control.
- Outside cells allowed with penalty.
- Used in most designs.

(C) PARTIAL REGION BOUND



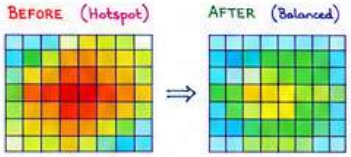
- Half or partial area bounded.
- Inside preferred.
- Good for hierarchical partitioning.

8. HOW REGION BOUNDS ARE USED ?



9. OPTIMIZATION USING REGION BOUNDS

- ✓ Adjust region shape / size to balance density.
- ✓ Tune Max Density (%) and Min Density (%) values.
- ✓ Use density heat map to identify hotspots.
- ✓ Overlapping regions with different priorities if needed.
- ✓ Re-run placement after adjustments.
- ✓ Ensure routing channels remain free.



Density Range (%):

- 0 - 20 (Low)
- 20 - 40 (OK)
- 40 - 60 (High)
- 60 - 80 (Very High)
- > 80 (Critical)

10. REGION BOUNDS VS OTHER CONSTRAINTS

Constraint Type	Blocks Cells Completely ?	Density Control	Routing Allowed ?	Typical Use
Hard Blockage	Yes	Not Applicable	No	Macros, IPs, Keep-outs, DRC
Soft Blockage	No	Yes (Penalty)	Limited	Density control, Congestion relief
Fence (Soft)	No	Yes (Preferred)	Yes	Guide cells within boundary
Region Bounds	No	Yes (Density Based)	Yes	Control density & distribution

11. INTERVIEW / VIVA Q & A

Q. What is region bound ?
→ It soft constraint used to control placement density in a specific area.

Q. Is region bound a hard blockage ?
→ No, it does not block cells completely.

Q. Why do we use region bounds ?
→ To guide placement, control congestion, reserve spaces and improve timing.

Q. Can cells go outside region bound ?
→ Yes, in soft bounds but with density penalty or lower priority.

12. SUMMARY POINTS

- ✓ Region bounds control where cells can be placed.
- ✓ It is a soft, density based constraint.
- ✓ Helps in density control and congestion reduction.
- ✓ Three types: Hard, Soft and Partial.
- ✓ Used for area reservation, ECO, timing and hierarchical partitioning.

13. CONGESTION SEVERITY (Po BASED)

- Low : 0% - 40% (Good)
- OK : 40% - 70% (Normal)
- Elevated : 70% - 100% (Watch)
- High : 100% - 150% (Critical)
- Severe : > 150% (Routing Risk)

QUICK RECAP FLOW



PLACEMENT

PLACEMENT BOUNDS (HARD / SOFT / PARTIAL)

1. DEFINITION

- Placement Bounds define the legal area within which cells are allowed to be placed during placement.
- Bounds control cell distribution, density and help meet design intent and constraints.
- Bounds do not completely block placement, but restrict cells to stay inside the region.

> Bounds = Soft constraint that guides cells to stay inside a defined region.

2. WHY PLACEMENT BOUNDS ARE USED ?

- Control density and avoid over-congestion.
- Reserve space for ECO / decap / fillers / memories.
- Group related logic together for performance.
- Keep high speed or sensitive blocks in small region.
- Hierarchical partitioning and area planning.
- Improve timing by reducing long interconnects.
- Support multi-power / multi-voltage partitioning.

3. CHARACTERISTICS

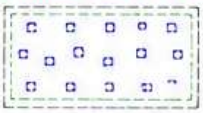
- ✓ Soft constraint (not a blockage).
- ✓ Cells allowed inside with density limits.
- ✓ Can be rectangular, polygonal or user defined shape.
- ✓ Has Max Density (%) and Min Density (%) limits.
- ✓ Overlapping bounds allowed with different priorities.
- ✓ Useful in floorplan based placement flow.

4. EFFECTS / BENEFITS

- Ensures balanced cell distribution.
- Reduces congestion and routing detours.
- Improves timing by clustering logic.
- Reserves space for future ECO / decap.
- Improves routability and IR drop performance.
- Maintains design intent and partitioning.

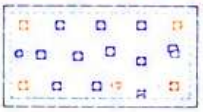
5. TYPES OF PLACEMENT BOUNDS

(A) HARD BOUND (STRICT BOUND)



- ✓ Cells MUST be placed inside.
- ✓ No cells allowed outside.
- ✓ Used for strict partitioning.
- ✓ Common for SRAM / PLL region.

(B) SOFT BOUND (DENSITY BASED)



- ✓ Cells preferred inside region.
- ✓ Outside allowed with penalty.
- ✓ Controlled by Max / Min Density.
- ✓ Most widely used bound.

(C) PARTIAL BOUND (MIXED REGION)

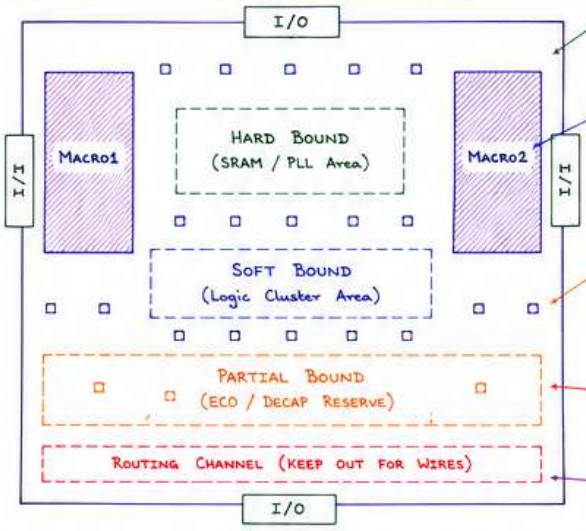


- ✓ Only portion of die is bounded.
- ✓ Cells preferred in bounded part.
- ✓ Outside area limited by density.
- ✓ Used for area reservation.

LEGEND :

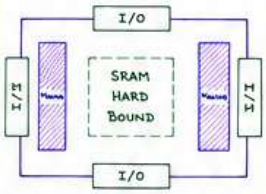
- = Standard Cell
- = Hard Bound (Strict)
- = Partial Bound Area
- = Macro / IP
- = Soft Bound (Density)
- = Cells Placed Outside
- = Routing Channel (Free)

6. PLACEMENT BOUNDS - DIAGRAM EXPLANATION



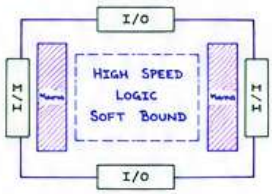
7. EXAMPLES OF PLACEMENT BOUNDS

(A) HARD BOUND EXAMPLE



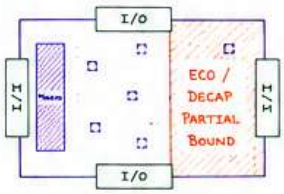
- Used for memories / PLL / analog.
- Strict partitioning required.
- No cells outside allowed.

(B) SOFT BOUND EXAMPLE



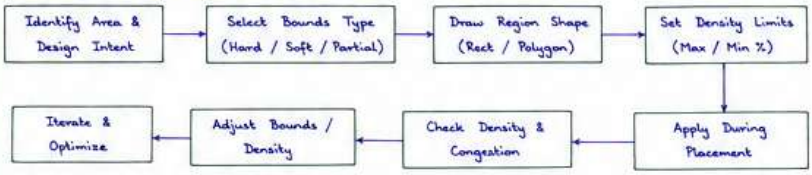
- High speed logic clustered.
- Density controlled (e.g. 30%).
- Outside cells allowed with density penalty.

(C) PARTIAL BOUND EXAMPLE



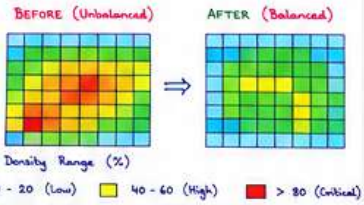
- Reserve only part of region.
- Used for ECO / decap space.
- Cells outside allowed with lower priority.

8. HOW TO DEFINE BOUNDS ?



9. OPTIMIZATION USING BOUNDS

- ✓ Tune Max Density (%) to balance congestion & timing.
- ✓ Reduce overlapping bounds to avoid conflicts.
- ✓ Use soft bounds for clustering and hard bounds for partitioning.
- ✓ Adjust region shape/size based on congestion heat map.
- ✓ Re-run placement after adjusting bounds.
- ✓ Ensure routing channels remain free between regions.



10. BOUNDS VS OTHER CONSTRAINTS

Constraint Type	Blocks Cells Completely ?	Density Control	Routing Allowed ?	Typical Use
Hard Blockage	Yes	Not Applicable	No	Macros, IOs, Keep-outs
Soft Blockage	No	Yes (Density based)	Limited	Density control, Conge. relief
Fence (Soft)	No	Yes (Preferred area)	Yes	Guide cells within boundary
Region Bounds	No	Yes (Density based)	Yes	Area partitioning, clustering

11. INTERVIEW / VIVA Q & A

Q. What are placement bounds ?
→ Soft constraints that guide cells to be placed inside a defined region.

Q. Difference between bounds and blockages ?
→ Blockages completely stop cells. Bounds allow cells with limits.

Q. Why soft bounds are useful ?
→ They help in clustering, density control and congestion reduction.

Q. What is max density in bounds ?
→ Maximum legal cell density allowed inside the bound region.

12. SUMMARY POINTS

- ✓ Bounds guide cell placement within regions.
- ✓ Types: Hard (strict), Soft (density), Partial.
- ✓ Control density and ensure better routability.
- ✓ Reserve areas for ECO, decap, memories.
- ✓ Improve timing by clustering related logic.
- ✓ Bounds are soft constraints, not blockages.
- ✓ Proper tuning leads to optimized PPA.

13. CONGESTION SEVERITY (Po BASED)

- Low : 0% - 40% (Good)
- OK : 40% - 70% (Normal)
- Elevated : 70% - 100% (Watch)
- High : 100% - 150% (Critical)
- Severe : > 150% (Routing Risk)

QUICK RECAP FLOW



PLACEMENT

HIGH FANOUT NET

1. Definition

- A High Fanout Net (HFN) is a net that drives a very large number of loads (sinks).
- Fanout = Number of end points connected to the net.
- When fanout count is higher than a technology / tool threshold (e.g., > 50, 100 or more), it is treated as a high fanout net.

Examples : Clock, Reset, Scan Enable, Power Control, Mode Select, Global Control signals etc.

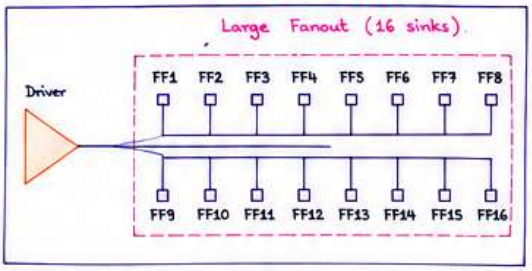
2. Examples

- Clock net driving thousands of flip-flops.
- Reset / Test enable driving all flops.
- Scan enable used in DFT.
- Power gating enable controlling many power switches.
- Global mode / boot signals used across the chip.

3. Why it is a Problem ?

- Large capacitive loading → high delay and slew.
- Causes significant signal skew (for functional nets).
- Increases dynamic and leakage power.
- Creates placement & routing congestion.
- Difficult to meet timing (setup / hold).
- May cause IR drop due to large switching activity.
- Makes noise and signal integrity worse (EM / antenna risks).

4. Diagram - Problem Without Optimization



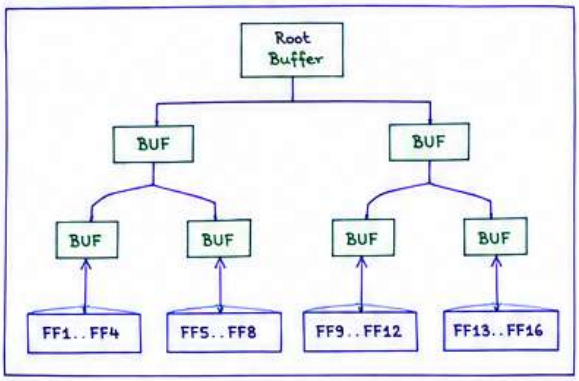
Problems :

- High load
- High delay
- High power
- Skew
- Congestion

5. Solutions for Problems

✓ Buffer Insertion / Fanout Splitting	→ Divide the net using buffer tree.
✓ Clock Tree Synthesis	→ For clock nets, build balanced clock tree with controlled skew.
✓ Hierarchical Distribution	→ Use local buffers for different blocks / regions.
✓ Shielding / Wider Metal	→ Route on higher metal with shielding for critical nets.
✓ Placement Clustering	→ Keep related sinks closer.
✓ Use of Repeater Cells	→ Improve slew and reduce RC delay.
✓ Logical / Functional Partitioning	→ Divide design to reduce fanout inside each partition.

6. Diagram - Optimized using Buffer Tree / CTS



- ✓ Load divided
- ✓ Lower delay
- ✓ Controlled skew
- ✓ Lower power
- ✓ Better routing

7. Types / Cases of High Fanout Nets

Type	Typical Fanout	Example	Handling Method
Clock Net	1000s	Clock to all FFs	CTS (Clock Tree)
Reset Net	100s - 1000s	Global reset	Buffer tree / Replica
Scan Enable	100s - 1000s	DFT scan_en	Hierarchical buffers
Mode / Config	100s	operating mode	Fanout splitting

8. Placement Guidelines

- Keep driver near geometric center for global nets (e.g., clock).
- Place sinks in clustered / balanced manner.
- Avoid congestion around high fanout drivers.
- Reserve space for buffers inserted during optimization.
- Use halos / keepouts to protect clock drivers.
- Prefer higher metal layers for routing high fanout nets.

9. Advantages

- ✓ Proper handling improves timing.
- ✓ Reduces power by balanced distribution.
- ✓ Minimizes routing congestion.
- ✓ Improves signal integrity.
- ✓ Helps in predictable and reliable design.

10. Disadvantages

- → Harder to meet timing.
- → Increases area due to buffers.
- → Extra power due to buffer tree.
- → Congestion if not handled well.
- → More design and routing complexity.

11. Before vs After Optimization

Parameter	Before (High Fanout)	After (Buffer Tree / CTS)
Load Capacitance	Very High	Split & Reduced
Net Delay	High	Lower
Transition Time	Large	Controlled
Signal Skew	More	Minimized
Dynamic Power	High	Optimized
Routing Congestion	High	Reduced
Timing Closure	Difficult	Achievable

12. Interview / Viva Q & A

Q. What is a high fanout net ?
→ A net that drives very large number of loads.

Q. Why it is a problem ?
→ High load, delay, power and congestion.

Q. How to fix it ?
→ Insert buffers, use CTS, cluster sinks, use higher metal, partition the design.

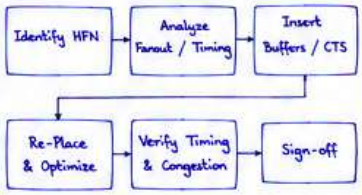
13. Summary Points

- ✓ High fanout nets have very large load.
- ✓ Causes delay, power and congestion.
- ✓ Common examples : clock, reset, scan_en.
- ✓ Handled using buffer tree / CTS.
- ✓ Proper placement and clustering is important.
- ✓ Optimization improves timing and reliability.

14. Key Point

High fanout nets must be handled properly using buffer insertion / CTS to achieve better timing, lower power and reliable physical design closure.

Quick Recap Flow



PD PLACEMENT – LEGALIZATION OF PLACEMENT

1. DEFINITION

- Legalization (legal placement) is the process of moving cells from the global placement result to legal, manufacturable locations that satisfy all placement rules and design constraints.

2. GOALS

- Place every cell in a legal site location.
- Ensure cells lie within placement rows and core bounds.
- Prevent cell overlap and maintain row utilization.
- Minimize cell movement from global placement.
- Preserve timing, wirelength, congestion and density quality.
- Maintain relative ordering to reduce routing impact.

3. WHY LEGALIZATION IS NEEDED ?

- Global placement may place cells anywhere (continuous coordinates) ⇒ cells are not aligned to rows.
- Cells may overlap or violate site, row and block constraints.
- Tool must convert the placement to discrete legal sites so that routing can be performed physically.
- Without legalization ⇒ DRC violations, routing failures, timing degradation.
- Legal placement is mandatory before optimization, CTS and routing.

4. LEGALIZATION ALGORITHM

1) Row assignment : Assign each movable cell to row (site aligned).
2) Sorting : Sort cells in each row by x-coordinate (left ⇒ right).
3) Conflict removal : Resolve overlaps & spacing violations by shifting cells left or right.
4) Gap distribution : Distribute gaps to meet target density or to reduce wirelength.
5) Iteration (if needed) : Re-run to improve quality and reduce displacement.

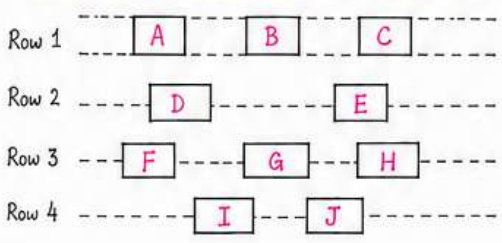
5. GLOBAL (UNLEGAL) vs LEGAL (LEGALIZED) PLACEMENT

PARAMETER	GLOBAL PLACEMENT (UNLEGAL)	LEGAL PLACEMENT (LEGALIZED)
Purpose	Optimize wirelength / timing using continuous model	Convert to legal, manufacturable locations
Location	Any (continuous coordinates)	Discrete legal sites on rows
Row alignment	Not aligned to rows	Aligned to placement rows
Overlap	Cells may overlap	No overlaps
Spacing	Spacing violations possible	All spacing rules satisfied
Constraints	May violate row / region / block constraints	All placement constraints satisfied
Quality metrics	Wirelength / timing / density optimized	Minimal displacement / row utilization / legality
Output	Unlegal placement	Legal placement
Used for	Global placement stage	CTS, Routing, Optimization

13. SUMMARY / REVISION POINTS

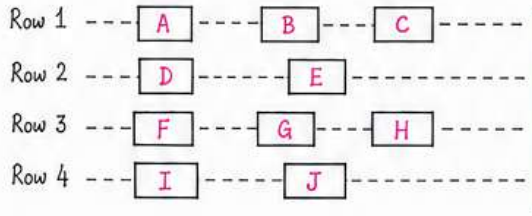
- Global placement uses continuous coordinates; cells are unaligned.
- Legalization aligns cells to rows and legal sites.
- Removes overlaps, resolves spacing violations and satisfies constraints.
- Uses row assignment, sorting, conflict removal, gap distribution and iteration.
- Legal placement is essential before CTS, routing and optimization.
- Good legalization quality leads to better PPA and routing success.

6. GLOBAL (UNLEGAL) PLACEMENT EXAMPLE



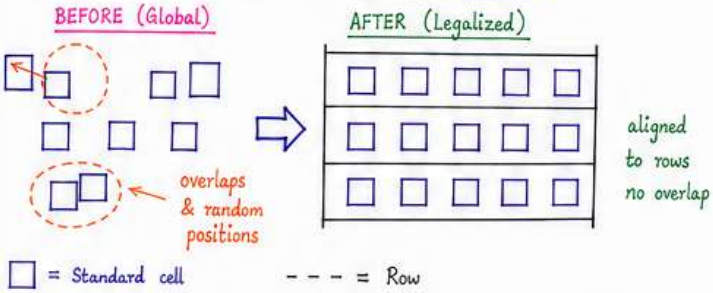
- Cells are not aligned to rows.
- Some cells overlap.
- Spacing violations exist.
- Not manufacturable.

7. LEGAL (LEGALIZED) PLACEMENT EXAMPLE



- All cells are aligned to rows.
- No overlap.
- Legal site locations.
- Ready for CTS and routing.

8. BEFORE vs AFTER LEGALIZATION (Illustration)



9. KEY POINT

Legalization converts the soft, continuous global placement into a hard, legal placement by aligning cells to rows, removing overlaps and satisfying all physical rules with minimal movement.

10. ADVANTAGES

- ✓ Produces legal, DRC clean placement.
- ✓ Enables successful routing.
- ✓ Maintains timing & wirelength quality.
- ✓ Improves row utilization and density.
- ✓ Foundation for CTS and timing closure.

11. PRACTICAL OBSERVATION

- Good global placement : cells may ⇒ small legalization displacement.
- If legalization movement ⇒ ~10-15% of design span ⇒ timing may degrade.
- Dense legalization constraints (fence / region / blockages) carefully.

12. INTERVIEW / VIVA POINTS

Q1. What is legalization in placement ?

⇒ It is the process of converting global placement to legal row-based placement by aligning cells to sites, removing overlaps and satisfying all placement rules.

Q2. Why is legalization important ?

⇒ Without legalization, cells may overlap or violate row and spacing rules, causing DRC violations and routing failures.

Q3. What happens if legalization moves cells too much ?

⇒ Excessive movement can degrade timing and wirelength, increase congestion and overall PPA quality worsens.

Author : Konda Tharun Vivekananda

PD PLACEMENT – ROW BASED LEGALIZATION

1. DEFINITION

- Row based legalization is the process of moving cells from global placement (continuous coordinates) to legal locations inside standard cell rows, such that all physical constraints and design rules are satisfied.

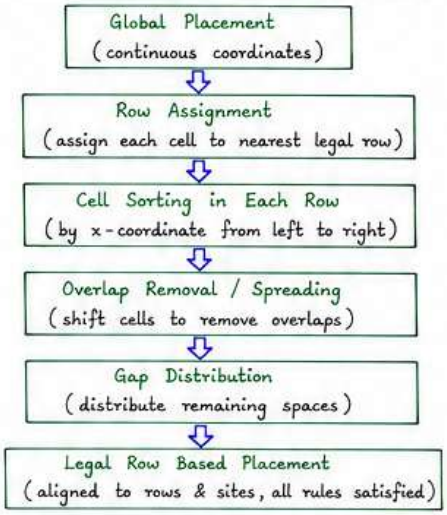
2. GOALS

- Align every cell to a row and site location.
- Remove overlaps and fix spacing violations.
- Minimize cell displacement from global placement.
- Preserve wirelength, timing and routability as much as possible.
- Produce a legal, manufacturable placement for next stages (CTS, routing, sign-off).

3. WHY VIS LEGALIZATION IS NEEDED ? (REASONS)

1) Global placement gives cell locations in continuous (x,y) coordinates. These locations do not follow row & site constraints.
2) Cells may overlap each other or blockages.
3) Cells may lie between rows or on forbidden locations.
4) Minimum spacing rules between cells may be violated.
5) Without legalization ⇒ DRC violations, poor routing, timing failure and physical verification fails.

4. ROW BASED LEGALIZATION – TYPICAL FLOW



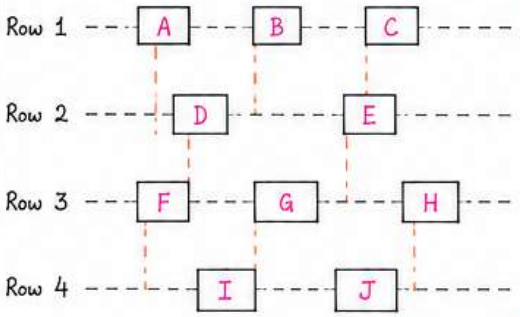
5. ADVANTAGES

- ✓ Produces legal placement aligned to rows and sites.
- ✓ Removes overlaps and spacing violations.
- ✓ Ensures DRC clean placement.
- ✓ Maintains good timing and wirelength as much as possible.
- ✓ Provides proper foundation for CTS, routing and sign-off.

6. DISADVANTAGES / CHALLENGES

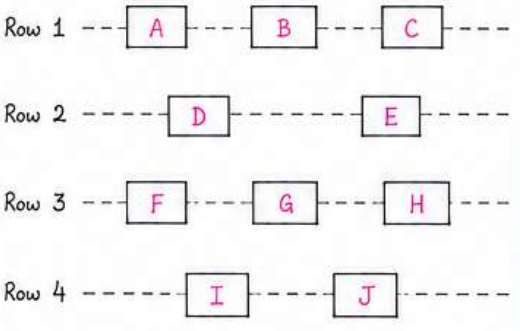
- × Requires cell movement ⇒ may increase wirelength.
- × Large displacement may degrade timing.
- × Over-constrained designs (dense blocks) may need large shifts.
- × Gap filling can create routing congestion.
- × Quality depends on row assignment and sorting heuristics.

7. EXAMPLE – GLOBAL PLACEMENT (CONTINUOUS COORDINATES)



- Cells are at continuous (x,y) locations.
- Not aligned to rows and sites.
- Overlaps and spacing violations may exist.

8. EXAMPLE – AFTER ROW BASED LEGALIZATION



- All cells aligned to rows.
- No overlaps.
- Minimum spacing satisfied.
- Placed on legal site locations.
- Legal placement for next stages.

□ = Standard Cell - - - = Row | = Site

9. GLOBAL (UNLEGAL) vs ROW BASED LEGAL PLACEMENT

PARAMETER	GLOBAL PLACEMENT (UNLEGAL)	ROW BASED LEGAL PLACEMENT
Coordinates	Continuous (x, y)	Discrete (aligned to rows & sites)
Row alignment	Not aligned	Aligned to rows
Overlap	May overlap	No overlap
Spacing	May violate spacing	Minimum spacing satisfied
DRC	May violate	DRC clean
Movement	Cells anywhere	Minimal movement to legal sites
Output	Soft solution	Hard, manufacturable solution
Used for	Next : Legalization	Next : CTS, Routing, Sign-off

10. PRACTICAL OBSERVATIONS

- Good global placement ⇒ small legalization displacement.
- If legalization movement ≥ ~10-15% of design span ⇒ timing may degrade.
- Dense placement (high utilization) ⇒ more gaps and congestion risk.
- Legalization quality strongly impacts CTS and routing success.

11. SUMMARY / KEY POINTS

- Row based legalization converts continuous global placement to legal row-and-site based placement.
- It aligns cells to rows, removes overlaps and satisfies spacing rules.
- It is mandatory before CTS, routing and sign-off.
- Helps to achieve DRC clean, timing friendly and routable placement.
- Good legalization quality improves overall Physical Design closure.

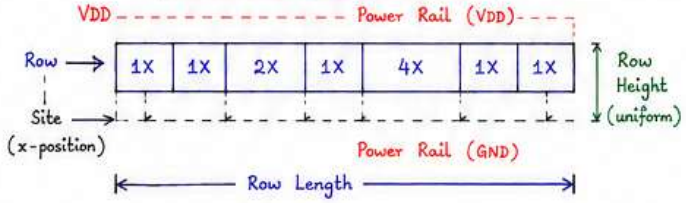
Author : Konda Tharun Vivekananda

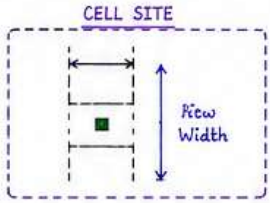
PD PLACEMENT – STANDARD CELL ARCHIECTURE & LEGALIZATION

1. STANDARD CELL ARCHITECTURE – DEFINITION

- Standard cell architecture defines the structure and rules used for designing and arranging standard cells in rows with uniform height to enable efficient placement, routing and timing closure.
- Cells are abutted in a row; power rails run horizontally.
- Cells of multiple widths (1X, 2X, 4X ...) but same height.

STANDARD CELL ROW STRUCTURE





Site : smallest legal placement unit.

= Cell contact (to power rails)

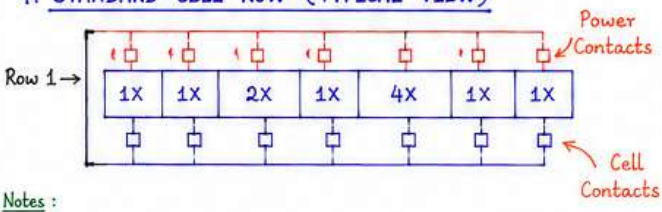
2. GOALS (OF STANDARD CELL ARCHITECTURE & LEGALIZATION)

1) Align all cells to legal rows and sites.
2) Remove overlaps and fix spacing violations.
3) Minimize cell movement from global placement.
4) Maintain timing, routability and wirelength quality.
5) Ensure DRC clean and manufacturable layout.
5) Provide a consistent structure for CTS, routing and sign-off.

3. WHY WE NEED LEGALIZATION ?

1) Global placement gives cells continuous (x,y) coordinates. Those locations are not aligned to rows and sites.
2) Cells may overlap or violate minimum spacing.
3) Cells may be placed in forbidden locations (fence / blockages).
4) Row height, site width and power rail connectivity will be violated.
5) Without legalization ⇒ DRC violations, routing congestion, timing degradation and sign-off failure.

4. STANDARD CELL ROW (TYPICAL VIEW)



Notes :

- ✓ Cells share power rails through contacts.
- ✓ All cells in a row have same height.
- ✓ Cells abut (no gap) except required spacing after legalization.

5. ADVANTAGES

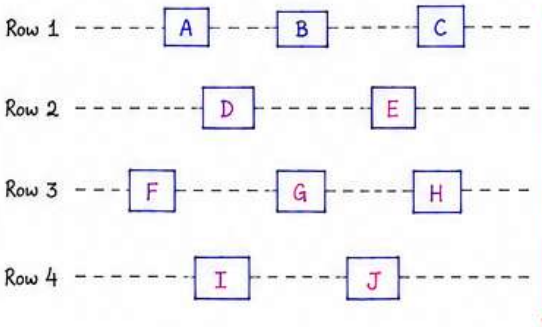
- ✓ Regular structure ⇒ easy placement, routing and verification.
- ✓ Better routability and less congestion.
- ✓ Maintains timing and wirelength quality.
- ✓ Ensures DRC clean and manufacturable layout.
- ✓ Simplifies CTS and power distribution.
- ✓ Scalable for large designs.

6. DISADVANTAGES / LIMITATIONS

- × Legalization movement may increase wirelength.
- × Timing may degrade if movement is large.
- × Large designs need more iterations for good quality.
- × Some dense blocks may cause legalization congestion.

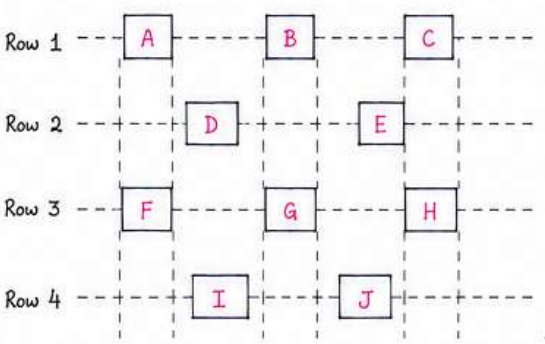
7. EXAMPLE – BEFORE vs AFTER LEGALIZATION

(a) BEFORE LEGALIZATION (Global Placement)



- Cells at continuous (x,y) locations.
- Not aligned to rows & sites.
- Overlaps may exist.
- Spacing violations may exist.

(b) AFTER LEGALIZATION (Row Based Legal Placement)



- All cells aligned to rows & sites.
- No overlaps.
- Minimum spacing satisfied.
- Legal placement ready for next stages.

= Standard Cell – – – = Row |– – –| = Site boundary.

8. SUMMARY – STANDARD CELL ARCHITECTURE & LEGALIZATION

ITEM	DESCRIPTION
Architecture	Cells of uniform height, multiple widths, arranged in rows with horizontal power rails.
Site	Basic placement unit (x-position) within a row.
Row	Horizontal track where cells are placed.
Legalization	Converts continuous global placement to row & site based legal placement.
Outcome	DRC clean, routable, timing friendly and manufacturable layout.
Next Stages	Optimization, CTS, Routing, sign-off.

9. PRACTICAL OBSERVATIONS

- Good global placement ⇒ small legalization displacement.
- Row utilization balance is important for legalization quality.
- Proper cell sorting reduces overlaps and spacing violations.
- Standard cell rows simplify power grid and routing.

10. KEY POINT

Standard cell architecture provides a regular, row based structure and legalization ensures all cells are aligned to rows and sites, removing overlaps and spacing violations for a legal, manufacturable physical design.

11. SUMMARY / REVISION POINTS

- Standard cell architecture ⇒ uniform row height, multiple widths.
- Power rails run horizontally, cells abut in rows.
- Legalization aligns cells to rows & sites.
- Removes overlaps, spacing violations and forbidden placements.
- Essential before CTS, routing and sign-off.
- Good legalization ⇒ better timing, routability and PPA.

Author : Konda Tharun Vivekananda

PD PLACEMENT – ALTERNATIVE ROW ORIENTATION & LEGALIZATION

1. DEFINITION

- Alternative row orientation allows adjacent rows to have opposite or different orientations (e.g., R0, MY, MX, R180) while keeping cells abutting and maintaining legal placement rules.
- Used during legalization to improve routability, reduce congestion and optimize wirelength.

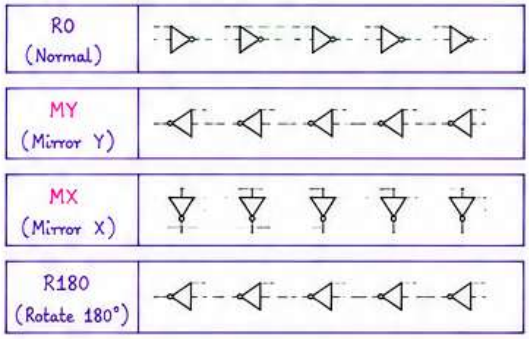
2. GOALS

- Improve routing flexibility by providing more routing tracks.
- Reduce congestion and wirelength.
- Maintain legal rows, sites and DRC rules.
- Enable better PPA (power, performance, area).
- Support timing closure by reducing detours.

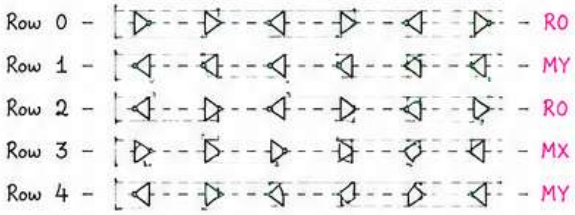
3. WHY WE NEED ALTERNATIVE ROW ORIENTATION ?

1) Standard same-orientation rows limit routing resources.
2) Long interconnects lead to congestion and timing degradation.
3) Some nets require vertical access ⇒ better with opposite row directions.
4) Helps to balance horizontal & vertical routing demand.
5) Improves placement density and reduces dead spaces.

4. ROW ORIENTATION TYPES



5. ALTERNATIVE ROW ORIENTATION PATTERN (EXAMPLE)



BENEFITS OF ALTERNATING ROWS

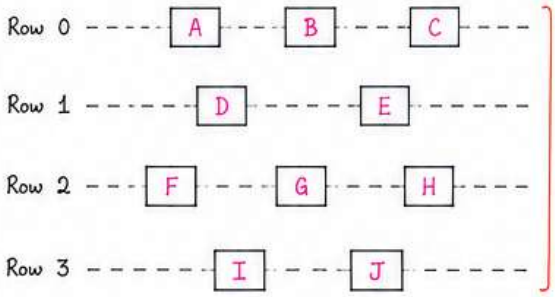
- ✓ Increases routing tracks in vertical direction.
- ✓ Provides access to power rails and signals from both sides.
- ✓ Reduces congestion in critical regions.

5. ADVANTAGES

- ✓ Better routability and higher routing resources.
- ✓ Reduced congestion and wirelength.
- ✓ Improved timing due to shorter interconnects.
- ✓ Efficient power distribution access.
- ✓ Higher placement density with less dead area.
- ✓ Supports complex designs and advanced nodes.

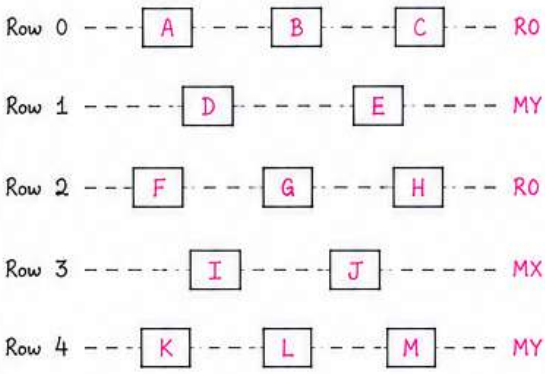
7. EXAMPLE – BEFORE vs AFTER LEGALIZATION

(a) BEFORE LEGALIZATION (Same Orientation)



- All rows same orientation.
- Limited routing tracks.
- Congestion possible.
- Long detours may happen.

(b) AFTER ALTERNATIVE ROW ORIENTATION



- Alternating row orientation.
- More routing tracks.
- Better access to power & signals.
- Reduced congestion & wirelength.

□ = Standard Cell – – – = Row | R0 / MY / MX / R180 = Row Orientation

8. COMPARISON – SAME ORIENTATION vs ALTERNATIVE ORIENTATION

PARAMETER	SAME ORIENTATION	ALTERNATIVE ORIENTATION
Routing Resources	Limited tracks (less access)	More tracks (better access)
Congestion	High in dense designs	Reduced significantly
Wirelength	Longer (more detours)	Shorter (optimized path)
Power Access	Single side access	Both side access
Timing	May degrade	Improved
Placement Density	Lower / uneven	Higher / uniform
DRC / Rules	Followed	Followed
Used for	Simple / small designs	Complex / large designs

9. DISADVANTAGES / LIMITATIONS

- × More complex floorplanning and row planning.
- × Increased legalizer runtime.
- × Power strap connections need careful planning.
- × May create alignment challenges in some macros.
- × More design iterations may be required.

10. PRACTICAL OBSERVATIONS

- Alternating rows is widely used in modern high-density ASICs.
- Improves routing efficiency and timing closure.
- Works best with proper power grid and pin access planning.
- Always verify DRC, power connectivity after legalization.

11. SUMMARY / KEY POINTS

- Alternative row orientation uses R0, MY, MX, R180 in adjacent rows.
- It increases routing tracks and reduces congestion.
- Enables better power access and shorter interconnects.
- Legalization ensures all cells are aligned to rows & sites.
- Improves timing, routability and overall PPA.
- Proper planning is essential to avoid DRC and alignment issues.

Author : Konda Tharun Vivekananda

PD PLACEMENT – OVERLAP REMOVAL & ITS IMPACT ON TIMING

1. DEFINITION

- Overlap occurs in global placement when two or more cells occupy the same or overlapping region of a row.
- Overlap removal (spreading) is the process of shifting cells to eliminate overlaps while keeping cells within legal rows and sites.

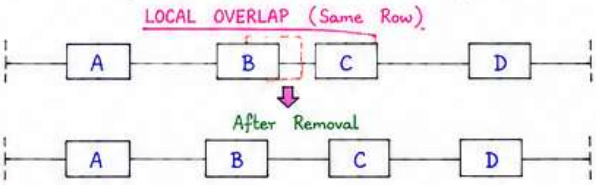
2. OVERLAP RESOLVING (HOW IT WORKS)

1) Detect overlapping cells in each row using cell positions and widths.
2) Build constraint graph (cells as nodes, overlaps as edges).
3) Choose a seed cell (usually with highest connectivity or largest fanout).
4) Move cells left or right to remove overlaps and legalize positions to nearest sites.
5) Update positions and repeat until no overlaps remain.
6) Ensure cells stay within row boundaries and DRC rules.
7) Recompute cost (wirelength, congestion) and iterate if required.

3. TYPES OF OVERLAP

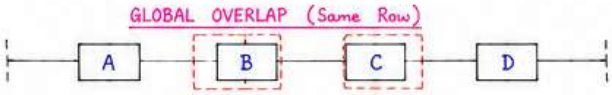
(a) LOCAL OVERLAP

- Overlap between cells that are close to each other in the same row.
- Usually due to small placement errors.
- Resolved by small shifts within local neighborhood.



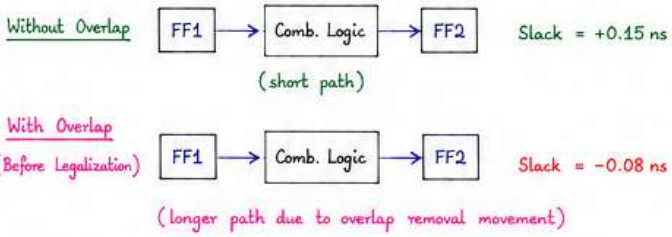
(b) GLOBAL OVERLAP

- Overlap between cells that are far apart or large groups in the row.
- Usually caused by high cell density or poor initial placement.
- May require long distance movement and multiple iterations.



4. IMPACT OF OVERLAP ON TIMING

- Overlaps create illegal placement ⇒ DRC violations.
- Cells forced to shift large distances during legalization ⇒ increase wirelength.
- Increased wirelength ⇒ higher RC delay.
- Higher RC delay ⇒ timing degradation (setup slack ↓, hold slack may also reduce).
- Can increase congestion ⇒ detours ⇒ more delay.
- May cause timing violations, routing failures and longer convergence to closure.

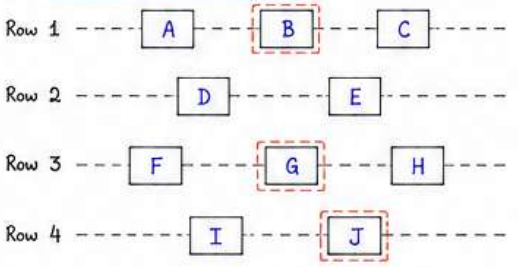


KEY POINT

Overlap removal is critical to obtain a legal placement, reduce congestion, control wirelength and achieve timing closure.

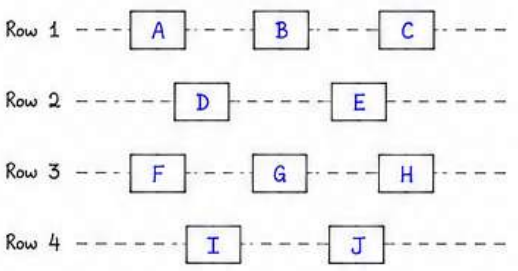
5. OVERLAP REMOVAL – VISUAL EXAMPLE (ROW BASED)

(a) BEFORE OVERLAP REMOVAL



- Multiple overlaps in each row.
- Not legal.
- May cause DRC violations, congestion and timing issues.

(b) AFTER OVERLAP REMOVAL (LEGALIZED)



- No overlaps.
- Cells aligned to sites.
- Ready for next stages (CTS, routing, etc).

□ = Standard Cell --- = Row | = Row boundary

⬚ = Overlap region

6. OVERLAP REMOVAL TECHNIQUES

1) Left-Right Algorithm : Sweep row left to right, push cells to right on overlap.
2) Right-Left Algorithm : Sweep row right to left.
3) Bidirectional / Iterative : Alternate L→R and R→L until no overlap.
4) Constraint Graph Spreading : Use graph to minimize total movement.
5) Cell Padding : Add spacing to prevent future overlaps.

7. COMPARISON – LOCAL vs GLOBAL OVERLAP

PARAMETER	LOCAL OVERLAP	GLOBAL OVERLAP
Location	Near cells in same region	Far apart or across row
Cause	Small placement error	High density / poor placement
Movement	Small shifts	Large shifts
Resolution Time	Fast	Slow / iterative
Impact on WL	Low	High
Impact on Timing	Minor	Significant
Complexity	Low	High

8. ADVANTAGES

- ✓ Produces legal placement.
- ✓ Removes DRC violations.
- ✓ Reduces wirelength and congestion.
- ✓ Improves routability.
- ✓ Helps achieve timing closure.

9. DISADVANTAGES / LIMITATIONS

- × Large movement may increase WL.
- × May degrade timing if excessive shifts.
- × Iterative process ⇒ more runtime.
- × Over-constrained areas may still cause congestion.

10. PRACTICAL OBSERVATIONS

- Good initial placement ⇒ less overlap ⇒ faster legalization.
- Overlap removal quality) strongly affects CTS and routing success.
- Dense blocks / macros may create global overlaps.
- Better legalization ⇒ better PPA and sign-off quality.

11. SUMMARY / REVISION POINTS

- Overlap occurs when cells occupy the same region in a row.
- Overlap removal (spreading) fixes overlaps and legalizes placement.
- Types : Local overlap (near cells) and Global overlap (far cells).
- Overlap legalizeases ⇒ [illegible], th, congestion and timing delay.
- Removal uses sweep or graph based algorithms to shift cells.
- Good overlap removal ⇒ legal, routable and timing friendly design.

Author : Konda Tharun Vivekananda

PD PLACEMENT – TIMING AWARE LEGALIZATION

1. TIMING AWARE LEGALIZATION – DEFINITION

- Timing aware legalization is the process of converting global placement to legal locations while considering timing criticality of nets and cells.
- It moves cells within rows to legal sites while trying to minimize timing degradation and fix violations.
- Goal is to obtain a legal placement with minimal impact on setup / hold timing and better overall slack.

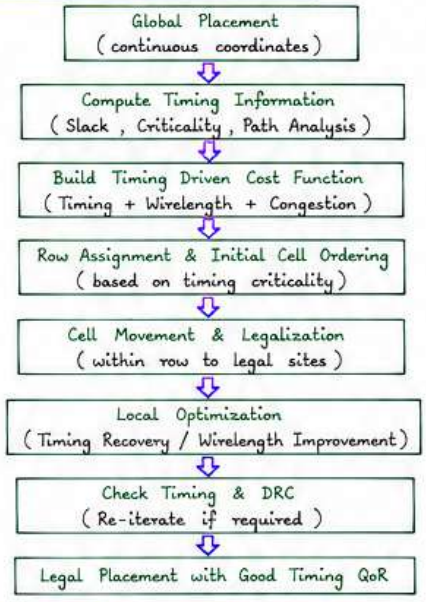
2. GOALS

- Preserve or improve timing (setup slack ↑, hold slack OK).
- Minimize wirelength while legalizing to sites.
- Prioritize timing critical paths during cell movements.
- Avoid unnecessary detours and congestion increase.
- Meet all DRC rules and row / site constraints.

3. WHY TIMING AWARE LEGALIZATION IS NEEDED ?

1) Global placement may place cells at continuous (x,y) locations that are illegal and may break row / site rules.
2) Simple legalization without timing considers only legality and congestion, which can degrade timing.
3) Critical path cells may be displaced far away, increasing net delay and reducing slack.
4) Timing aware approach reduces timing QoR loss and helps achieve timing closure faster.

4. TIMING AWARE LEGALIZATION FLOW



5. HOW TIMING IS CONSIDERED ?

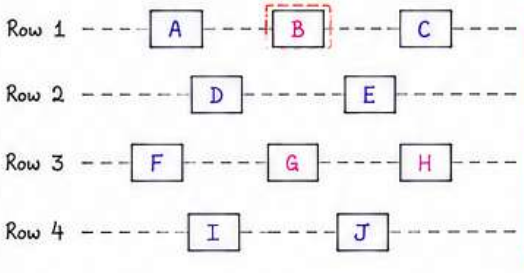
- Each cell / net is given a criticality value based on timing paths.
- Cells on critical paths are moved less aggressively.
- Cost function = α x Timing Cost + β x Wirelength Cost + γ x Congestion Cost
- Timing cost $\propto$ (Criticality x Slack Sensitivity)

KEY POINT

Timing aware legalization balances legality with timing, so that critical paths are protected and overall design QoR improves.

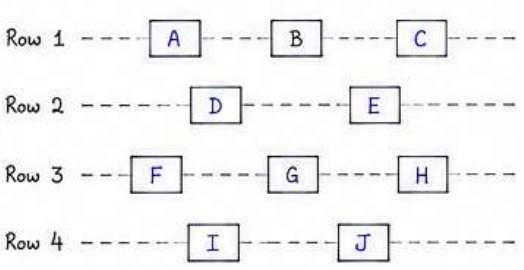
6. EXAMPLE – TIMING AWARE LEGALIZATION

(a) BEFORE LEGALIZATION (GLOBAL PLACEMENT)



- Cells at continuous locations.
- Overlaps exist (B with neighbors, G with neighbors).
- May create timing degradation.

(b) AFTER TIMING AWARE LEGALIZATION



- All cells aligned to legal sites.
- Overlaps removed.
- Critical cells kept near original positions.
- Timing protected.

□ = Standard Cell – – – = Row | = Site boundary

7. TYPES OF TIMING CONSIDERATION

(a) SLACK BASED

- Cells with negative / low slack get higher priority.
- Legalization tries to minimize additional delay on these cells.

(b) CRITICALITY BASED

- Path criticality (0 to 1) computed from STA.
- High criticality cells are moved less to avoid timing loss.

(c) PATH BASED

- Preserve placement of entire critical path as much as possible.
- Legalize non-critical cells more aggressively.

8. IMPACT ON TIMING

SCENARIO	EFFECT ON TIMING
Without Timing Aware	Can increase path delay, reduce setup slack, introduce hold issues.
With Timing Aware	Minimizes delay increase, preserves setup slack, reduces hold violations.
Critical Path Cells Stable	Less delay change, better slack.
Non-Critical Cells Moved	More freedom for legalization, small timing impact.

9. ADVANTAGES

- ✓ Protects timing critical paths.
- ✓ Reduces timing degradation after legalization.
- ✓ Helps achieve timing closure with fewer iterations.
- ✓ Better overall QoR (PPA + timing).
- ✓ Improves routability without hurting timing

10. DISADVANTAGES / LIMITATIONS

- × Higher runtime due to timing analysis.
- × More complex cost function.
- × May limit cell movement freedom.
- × Dependence on accurate timing estimation.
- × Wrong weights ⇒ sub-optimal results.

11. PRACTICAL OBSERVATIONS

- Weight tuning between timing, wirelength and congestion is very important.
- Overly strict timing constraints during legalization may increase congestion.
- Good timing estimation before legalization gives better results.
- Works well with incremental STA updates.

12. SUMMARY / REVISION POINTS

- Timing aware legalization legalizes placement while protecting timing.
- Uses timing information (slack / criticality) in cost function.
- Critical cells are moved less, non-critical cells have more freedom.
- Helps to maintain setup slack, avoid hold violations.
- Reduces timing degradation and improves timing closure efficiency.
- Essential step before CTS and routing for better physical design QoR.

Author : Konda Tharun Vivekananda

PD PLACEMENT – POST PLACEMENT OPTIMIZATION

1. INTRODUCTION

- Post placement optimization (PPO) is performed after legal placement and before CTS to improve timing, power, congestion and routability without violating legality.
- It makes small controlled adjustments and/or netlist changes to achieve better QoR.

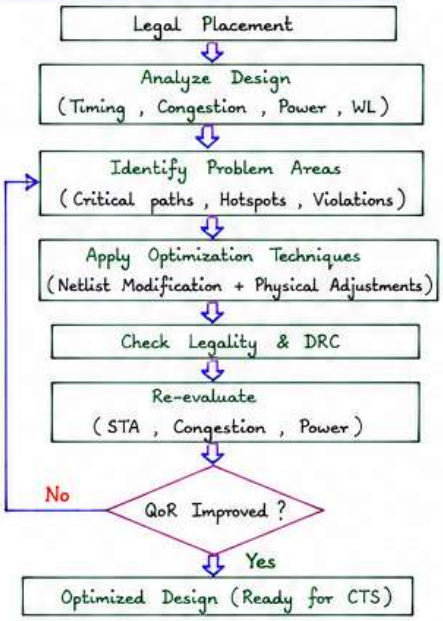
2. WHY POST PLACEMENT OPTIMIZATION ?

- Initial placement may still have timing critical paths.
- To reduce wirelength and congestion hotspots.
- To fix local timing violations (setup / hold).
- To improve power efficiency by reducing switching and capacitive load.
- To prepare a better design for CTS and routing.

3. GOALS

- ✓ Improve timing slack (setup ↑, hold OK).
- ✓ Reduce total wirelength and congestion.
- ✓ Remove remaining overlaps / spacing issues.
- ✓ Balance delay across the design.
- ✓ Maintain legality and DRC rules.
- ✓ Improve power, area and overall QoR.

4. POST PLACEMENT OPTIMIZATION FLOW



5. OPTIMIZATION TECHNIQUES

(a) NETLIST MODIFICATION

- Cell sizing (upsize / downsize).
- Buffer insertion on critical nets.
- Gate cloning / duplication.
- Logic restructuring / restructuring.
- Constant propagation / logic simplification.
- Use of alternative equivalent cells.

(b) PHYSICAL ADJUSTMENTS

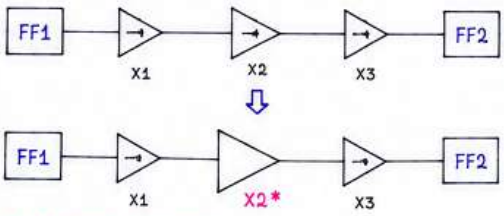
- Cell movement (within row / neighboring rows).
- Cell swapping.
- Cell spreading for congestion relief.
- Reordering of cells in a row.
- Macro / block relocation (if allowed).
- Spacing adjustment to improve routability.

KEY POINT

Post placement optimization fine tunes the design after legalization to achieve better timing, power, congestion and routability before CTS.

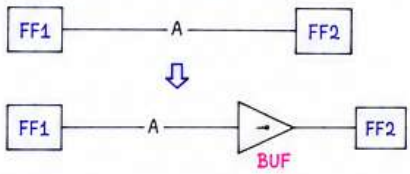
6. EXAMPLES OF OPTIMIZATION

(a) CELL RESIZING



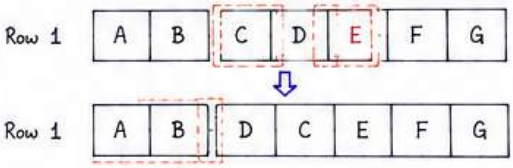
- Upsize X2 to reduce delay on critical path.

(b) BUFFER INSERTION



- Insert buffer to reduce load and improve delay.

(c) CELL MOVEMENT (PHYSICAL ADJUSTMENT)



- Move / swap cells to shorten nets and fix local timing.

□ = Standard Cell – – – = Row ⬚ = Problem Area

7. BEFORE vs AFTER OPTIMIZATION (TIMING SLACK)

(a) BEFORE OPTIMIZATION

Path	Slack (ns)
Path 1 (critical)	– 0.25
Path 2	+ 0.10
Path 3	– 0.05

(b) AFTER OPTIMIZATION

Path	Slack (ns)
Path 1 (critical)	+ 0.12
Path 2	+ 0.18
Path 3	+ 0.05

Negative slack (red) → improved to positive slack (green)

8. IMPACT ON DESIGN

METRIC	IMPACT AFTER PPO	COMMENTS
Timing	Improved slack	Critical paths optimized
Wirelength	Reduced	Shorter connections
Congestion	Reduced	Fewer hotspots
Power	Reduced	Lower cap load, less switching
Area	Slight increase / same	Due to buffers / upsizing
Routability	Improved	Easier routing
QoR	Better	Overall design quality improved

9. ADVANTAGES

- ✓ Better timing and power.
- ✓ Reduced congestion.
- ✓ Improved routability.
- ✓ Prepares design for CTS.
- ✓ Enhances overall QoR.

10. DISADVANTAGES / LIMITATIONS

- × May increase area and cell count.
- × More runtime / iterations.
- × Over optimization may cause unnecessary power.
- × Depends on quality of initial placement.

11. PRACTICAL OBSERVATIONS

- Small movements and local fixes give good results.
- Over optimization leads to diminishing returns.
- Balance between timing, power and congestion is important.
- Always verify legality after each optimization step.

12. SUMMARY / KEY POINTS

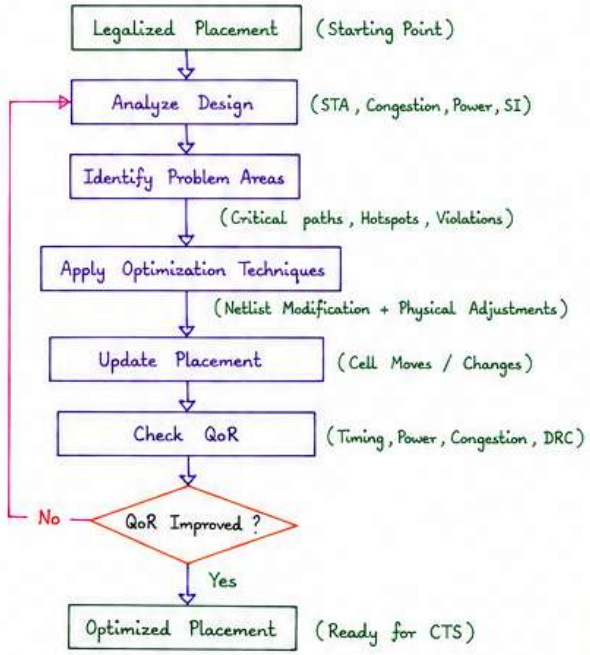
- PPO is performed after legal placement and before CTS.
- It uses netlist modifications and physical adjustments.
- Goals ⇒ better timing, lower power, lower congestion.
- Techniques ⇒ resizing, buffering, cloning, cell movement.
- Leads to improved QoR and smoother CTS & routing.

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PD PLACEMENT - POST PLACEMENT OPTIMIZATION

1. DETAILED PPO LOOP

- PPO is an iterative loop after legalization and before CTS to improve timing , power , congestion and routability .



- The loop continues until QoR meets the targets or no more significant improvement is observed .

2. PPO TARGETS

- ✓ Improve setup and hold timing slack .
- ✓ Reduce total wirelength and congestion .
- ✓ Remove remaining overlaps / spacing issues .
- ✓ Reduce switching power and leakage power .
- ✓ Balance delay across the design .
- ✓ Improve routability and signal integrity .
- ✓ Meet all DRC , LVS and row / site constraints .
- ✓ Achieve better overall QoR (quality of results) .

3. OPTIMIZATION TECHNIQUES

(a) NETLIST MODIFICATION (LOGICAL CHANGES)

- Cell sizing (upsizing / downsizing) to fix timing .
- Buffer insertion on critical nets .
- Gate cloning / duplication for high fanout .
- Logic restructuring / resynthesis of small cones .
- Constant propagation & logic simplification .
- Use of alternative / equivalent cells .
- Multi - Vt optimization for power & timing trade-off .

(b) PHYSICAL ADJUSTMENTS (PLACEMENT CHANGES)

- Cell movement within row / neighbouring rows .
- Cell swapping to reduce local congestion .
- Cell spreading on critical nets .
- Reordering cells in a row to reduce net delay .
- Macro / block relocation (if allowed by floorplan) .
- Density balancing to improve routability .
- Pin access optimization for better routing .
- Spacing adjustment for timing & DRC closure .

KEY POINT

PPO fine tunes the design after placement to achieve better timing , power , congestion and routability before CTS .

4. PPO TARGET METRICS (WHAT WE IMPROVE)

- Setup Slack
- Hold Slack
- Total Negative Slack (TNS)
- Worst Negative Slack (WNS)
- Total Wirelength
- Congestion
- Switching Power
- Leakage Power
- Cell Density
- DRC / Spacing Violations

GOAL

Improve these metrics without violating legality & constraints .

5. EXAMPLE - PPO IMPROVEMENTS

METRIC	BEFORE PPO	AFTER PPO	IMPROVEMENT
Worst Negative Slack (ns)	-0.22	+0.08	↑ 0.30 ns
Total Negative Slack (ns)	-12.40	-2.10	↑ 10.30 ns
Total Wirelength (μm)	12.5 M	10.2 M	↓ 18.4 %
Congestion (%)	76.5	58.2	↓ 18.3 %
Switching Power (mW)	128.7	114.3	↓ 11.2 %
DRC Violations	34	0	Eliminated

6. ADVANTAGES OF PPO

- ✓ Improves timing slack and reduces violations.
- ✓ Reduces wirelength and congestion .
- ✓ Improves power efficiency .
- ✓ Enhances routability and placement quality.
- ✓ Prepares design better for CTS and routing.
- ✓ Leads to better final QoR and fewer iterations later .

7. DISADVANTAGES / LIMITATIONS

- ✗ Increases runtime & tool effort.
- ✗ More iterations increase design turnaround time .
- ✗ May cause power increase due to upsizing / buffers .
- ✗ Excessive moves may cause placement thrashing .
- ✗ Depends on initial placement quality .

8. PLACEMENT VERIFICATION CHECKS AFTER PLACEMENT

CHECK	DESCRIPTION	TOOL / METHOD	PASS CRITERIA
Row / Site Legalization	All cells placed in legal rows & sites	Placement Tool	100% legal
Overlap Check	No cell overlaps	DRC / Verify	0 overlaps
Spacing Check	Min spacing between cells / sites	DRC / Verify	0 spacing violations
DRC Check	Physical design rule compliance	DRC	0 violations
Congestion Check	Routing demand vs capacity	Congestion Analyzer	Within target (e.g , < 70%)
Wirelength Check	Total wirelength estimation	Placement Tool	As low as possible
Timing Check (STA)	Setup / Hold slack verification	STA	Slack ≥ 0
Power Estimation	Est. switching & leakage power	Power Analyzer	Within target
Density Check	Cell density distribution	Density Tool	Uniform density
Pin Access Check	Pin accessibility for routing	Pin Access Analyzer	Meets target

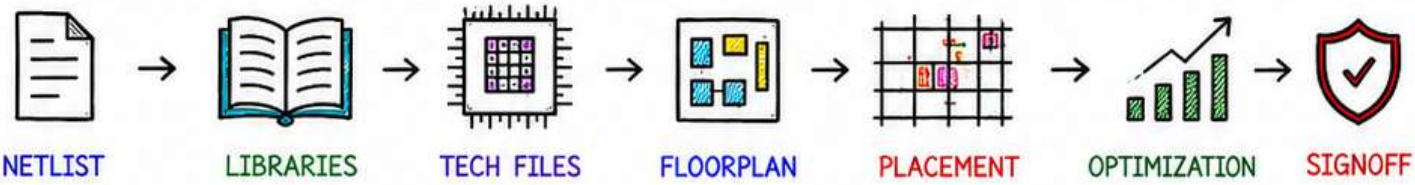
9. PRACTICAL OBSERVATIONS

- Good PPO reduces iterations in CTS and routing .
- Critical path cells are optimized first.
- Balance is required between timing & power .
- Over optimization may lead to oscillations .
- Consistent DRC clean placement eases sign-off .

10. SUMMARY / REVISION POINTS

- PPO is done after placement and before CTS.
- It improves timing , power , congestion & routability .
- Uses netlist modifications and physical adjustments .
- Goal is to meet all targets with good QoR.
- Verification checks ensure legal and clean placement .
- Proper PPO leads to smooth CTS & routing.

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NETLIST
- Gate Level Netlist
- Libraries
- Constraints

LIBRARIES

TECH FILES
- LEF
- DEF

FLOORPLAN

PLACEMENT

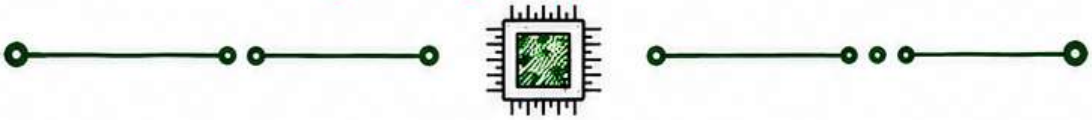
OPTIMIZATION

SIGNOFF
- DRC
- LVS
- Antenna
- Timing

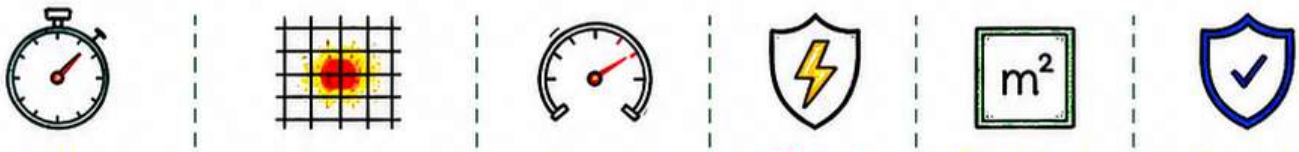


THANK YOU!

Your time and attention are
— greatly appreciated. —

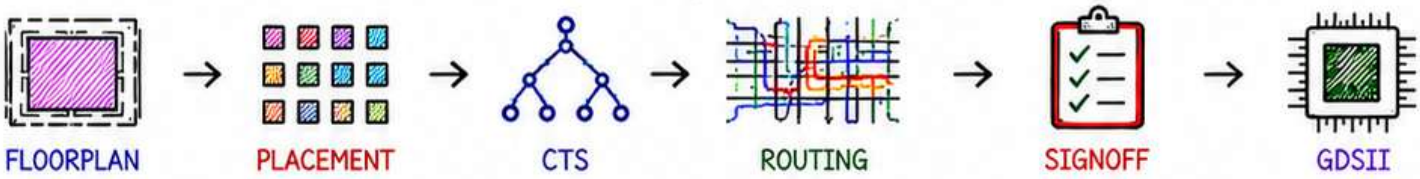


> Great Placement + Smart Optimization
> = Exceptional Chip.



Better Performance | Lower Congestion | Improved Timing | Efficient Power | Optimized Area | Signoff Ready

Thank you for being a part of this learning journey.
Keep Exploring, Keep Innovating, Keep Building the Future!



FLOORPLAN → PLACEMENT → CTS → ROUTING → SIGNOFF → GDSII

— || Let's continue to create better chips together! || —